71. Calculate the concentration of hydrogen ion in a solution that contains 0.01M $Ba(OH)_2$.

Solution

$Ba(OH)_2 \longrightarrow Ba^{2+} + 2OH^-$

1M 2M

0.01M ?

$\Rightarrow [OH^-] = 2 \times 0.01M = 0.02M$

From ion product of water, we know that:

$K_w = [H^+][OH^-]$

$$[O\bar{H}] = \frac{k_w}{[H^+]} or [H^+] = \frac{kw}{[OH^-]}$$

$$[H^+] = \frac{1\times10^{-14}M^2}{2\times10^{-2}M} = 5.0\times10^{-13}M$$

72. What is the POH of 0.02M H_2SO_4 solution?

Solution

$H_2SO_4 \longrightarrow 2H^+ + SO^2_4$

1 2M

0.02M ?

$\Rightarrow [H^+] = 2 \times 0.02M = 0.04M = 4\times10^{-2}M$

$P^{OH} = 14 + \log [H^+]$

$= 14 + \log 4 \times 10^{-2} = 12.6$

73. The concentration of OH^- ions in a certain hose hold ammonia cleaning solution is 0.0025M. calculate the concentration of the H^+ ions. Is the solution acidic, basic or neutral?

Solution

$$[H^+] = \frac{k_w}{[O\bar{H}]} = \frac{1\times10^{-14}M}{0.0025M} = 4.0\times10^{-12}M$$

$\Rightarrow [H]^+ < [O\bar{H}]$, the solution is basic.

74. What is the concentration of hydroxide ion of a solution whose POH value is 2?

Solution

$$[OH^-] = 10^{-POH} = 10^{-POH} = 10^{-2} = 0.01M$$

75. Determine the concentration of hydrogen ion when concentration of hydroxide ion is 10,000 times the concentration of hydrogen ion.

Solution

$$[OH^-] = 10000[H^+]$$

$$Kw = [OH^-][H^+]$$

$$1 \times 10^{-14}M^2 = 10000[H^+] \cdot [H^+]$$

$$[H^+][H^+] = \frac{1\times10^{-14}\,M^2}{10000}$$

$$[H^+] = \sqrt{1\times10^{-18}\,M^2}$$

$$[H^+] = 1.0\times10^{-9}\,M$$

Preparation of Bases

➦ Bases can be manufactured by

i **The reaction between base anhydride and water**

Base an hydride + water ⟶ base

Example

$Na_2O + H_2O \longrightarrow 2NaOH$

$MgO + H_2O \longrightarrow Mg(OH)_2$

ii **The reaction between active metals and water**

➦ This method is used to prepare soluble bases called **alkalis.**

Example

$2Na + 2H_2O \longrightarrow 2NaOH + H_2$

$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$

iii Double displacement reaction

➥ From soluble base and salt, another soluble base and insoluble salt are formed.

Example

$Ba(OH)_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2NaOH(aq)$

$Ca(OH)_2(aq) + K_2CO_3(aq) \longrightarrow 2KOH(aq) + CaCO_3(s)$

Sodium hydroxide (NaOH) and potassium hydroxide (KOH) are manufactured by electrolysis of concentrated sodium chloride and potassium chloride solution respectively.

Illustrative Examples

76. Which one of the following processes does NOT form a hydroxide base? (EGSECE....2001E.C)

A. Reaction of active metal with water

B. Reaction of alkali with a solution of salt

C. Direct combination of elements

D. Reaction of soluble metallic element with water

Explanation: Bases can be prepared by the reaction between active metals with water, basic anhydride with water and by double displacement reaction.

Answer C

77. Which of the following reactions forms a base?

A. Reaction of acidic oxide with water

B. Reaction of hydrogen with a non-metal

C. Reaction of active metals with water

D. Neutralization

Answer C

78. Complete and balance the following

a. $K + H_2O \longrightarrow$

b. $BaO + H_2O \longrightarrow$

c. $Ca(OH)_2 + Na_2CO_3 \longrightarrow$

Answer

a. $2K + 2H_2O \longrightarrow 2KOH + H_2$

b. $BaO + H_2O \longrightarrow Ba(OH)_2$

c. $Ca(OH)_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaOH$

79. What is the meaning of alkalis?

Explanation

Alkalis are soluble bases. Example NaOH, KOH, NH_3 etc

Uses of some important Acids and Bases

1. Sulphuric acid (H_2SO_4)

➦ It is rank number one important chemical in industry.

The major uses of sulphric acid are:

- ➦ To manufacture fertilizers, dyes, paints, explosives, drugs, rayon (artificial fibers), detergents.
- ➦ To clean metal surfaces
- ➦ Used as electrolytes in car batteries
- ➦ Used as oxidizing and dehydrating agents
- ➦ Used as catalyst in organic chemistry for example for making esters from alcohols and carboxylic acid

2. Hydrochloric acid (HCl)

It is used for:

- ➦ Cleaning metal surface before electroplating
- ➦ Preparation of drugs, dyes and photographic chemicals
- ➦ Preparation of chlorine gas

3. Nitric acid (HNO_3)

➦ Pure nitric acid is called " aqua fortis"

It is used to make fertilizers, explosives such as trinitro toluene (TNT) and to make aqua regia that dissolves gold.

4. Sodiumhydroxide (NaOH)

➦ It is commonly known as **caustic soda or lye.**

➦ It is used in:

- Primarly dry cells as electrolyte
- Making soap, pulp, paper and textile
- Purifying aluminum ores and removing sulphur from petroleium

5. Potassium hydroxide (KOH)

It is commonly known as **caustic potash.** It is stronger than sodiumhydroxide.

➦ It is used to make soft soaps and as electrolyte in alkaline batteries

6. Calcium hydroxide ($Ca(OH)_2$)

It is also known as **slacked lime**

➦ It is widely used to remove the acidity of soil.

➦ Used to remove temporarly hardness of water

➦ Used to remove hairs from hides in tanning process

➦ Used as a componenet of mortar (mixture of slacked lime, sand and water)

7. Ammonia (NH_3)

As solution of ammonia in water forms ammonium hydroxide.

It is used as:

➦ Household cleaning agent

➦ Laboratory reagent

➦ Raw material to make nitric acid and to make fertilizers

Precautions while working with acids and bases in the laboratory

- It is important to wear goggles, gloves and laboratory coat when dealing with acids and bases.
- Use bellows to pipette acidic or basic solutions rather than sucking it with your mouth
- If concentrated acid or corrosive base is spilled on cloth immediately wash with water.
- If acid enters your eye, wash it with water then consult doctor. If acid accidentally swallowed, administer weak bases such as magnesiumhydroxide ($Mg(OH)_2$) or aluminum hydroxide ($Al(OH)_3$)
- If a base is swallowed by accident, drink 1-2% acetic acid or lemone juice.
- Whenever bases come into contact with your skin, wash the affected part with water and than by weak acid solution such as acetic acid.
- During dilution of concentrated acid, add the acid into water (AW) but not water to acid (WA). Follow the Amharic AW! Not WA! (A-acid, W-water)

Illustrative Examples

80. Which of the following is used as a raw material in the production of nitric acid? (EGSECE)

A. $NaNO_3$ B. N_2O C. N_2O_3 D. NH_3

Explanation

Ammonia is used as a raw material to manufacture nitric acid acid industrially by Ostwald process.

Answer D

81. Which of the following is not true about ammonia?

A. Ammonia is used to make fertilizers.
B. Ammonia is used to as a cleaning agent.
C. Ammonia is used as anesthetic in dental and minor surgery.
D. Ammonia is used as refrigerant.

Answer: C

82. A country is said to be developed by its large quantity production and consumption of

A. Sulphuric acid
B. Nitric acid
C. phosphoric acid
D. Hydrochloric acid

***Explanation*:**

The development of a country can be determined by the amount of sulphuric acid it produces and consumes.

Answer A

83. One of the most important industrial use of nitric acid is

A. For dissolving some metals that cannot be dissolved by other acids.
B. For production of photographic films
C. For an oxiding agent
D. For the manufacture of fertilizers

Answer D

84. Which of the following is safe to perform in the laboratory?

A. Administering strong acid for swallowed base
B. Adding concentrated acid carefully to water in order to dilute
C. Washing an acid entered the eye with water then adding a weak base
D. Washing a splash concentrated acid on the body with water then treating with NaOH.

Answer B

85. Which of the following safety rules are not allowed in handling bases in high school laboratories?

A. Wearing eye goggles, gloves and laboratory coat
B. Washing the affected part with water whenever bases are in contact with our skin
C. Cleaning the working table immediately if bases are spilled on it
D. Tasting bases by using our sense organ tangue

Explanation

- Tasting chemicals is not allowed in laboratories even chewing gum or drinking water in laboratory is not permitted.

Answer D

7.3 Salts

- **Salts** are ionic compounds obtained by a neutralization reaction. They are formed when a metal ion or ammonium ion takes the place of hydrogen in an acid.
- The positive ion of a salt obtained from base and negative ion from acid.

Table 7.2. Acids and their salts name

Acids	Their salts
Sulphuric acid	Sulphates
Sulphurous acid	Sulphites
Nitric acid	Nitrates
Nitrous acid	Nitrites
Phospheric acid	Phosphates
Phosphorous acid	Phosphites
Perchloric acid	Perchlorates

Classification of salts

1. **Normal salts** – are formed when all the ionizable hydrogen of the acid replaced by metal ion or ammonium ion.

Example:

$NaCl$, NH_4NO_3, CaF_2 etc

2. **Acidic salts** – are slats that contains at least one unionized hydrogen.

Example:

- $KHCO_3$
- $(NH_4)_2HPO_4$
- $CaHPO_4$
- $Mg(HCO_3)_2$etc

3. **Basic salts** – are slats in which all the hydroxide ions of the base are not replaced

Example:

- $Pb(OH)Cl$
- $Cu(OH)NO_3$etc
- $Zn(OH)NO_3$

4. **Binary salts :** are formed from two types of elements only

Example:

$AlCl_3$, $CaBr_2$, KI, etc

5. **Ternary salts**: are formed from three types of elements.

Example:

$Al_2(SO_4)_3$, KNO_3, $NaClO_4$etc

Illustrative examples

86. A ternary salt among the following is

A. Al_2S_3
B. CH_3COONa
C. $Ca(SHO_3)_2$
D. $Ca(ClO_4)_2$

Answer: D

87. One of the following salts is acidic salt. Which one is it?

A. $Ca(NO_3)_2$
B. $Zn(OH)Cl$
C. KH_2PO_4
D. Na_2SO_4

Answer C

88. Identify the pair of compounds that is not related like the other pairs.

A. Nitric acid – sodium nitrate
B. Sulphuric acid – sulphurdioxide
C. Carbonic acid – calcium carbonate
D. Phosphoric acid – calcium phosphate

Answer B

89. Classify the following salts as normal, acidic, basic, binary or ternary salts

a. $LiHCO_3$
b. $BaSO_4$
c. K_2S
d. $Pb(OH)Cl$

e. NH_4Cl f. $CuCl_2$

Answer

- c and f are binary salts
- a is acidic salt
- b, c, e and f are normal salts
- b and e are ternary salts

90. write the name of the following salts

a. FeF_2 c. Li_3N e. $Zn(OH)NO_3$

b. $CuSO_4$ d. $Ca_3(PO_4)_2$ f. $NaHCO_3$

Explanation

- for binary salts the name of non-metal end with the suffix – ide. For ternary salts, the name of metal followed by the name of poly atomic ions.

a. Iron (II) fluoride d. calcium phosphate

b. copper (II) sulphate e. zinc basic nitrate

c. lithium nitride f. sodium hydrogen carbonate

91. why salts are called ionic compounds?

Explanation ➥

Because salts are aggregates of positive and negative ions

Preparation of salts

➥ There are different methods used to prepare salts.

No.	Method	Example
1	Direct combination of elements	$Ca + S \rightarrow CaS$ $3Mg + N_2 \rightarrow Mg_3N_2$
2	Reaction of active metals with acids	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
3	Reaction of basic oxide with acid	$Na_2O + H_2CO_3 \rightarrow Na_2CO_3 + H_2O$
4	Reaction of acid with base	$KOH+HCl \rightarrow KCl+H_2O$

5	Double displacement reaction	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
6	Reaction of carbonates with acid	$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + 2H_2O$
7	Reaction of hydrogencarbonates with acids	$NaHCO_3 + HNO_3 \rightarrow NaNO_3 + CO_2 + H_2O$

Illustrative examples

92. salts can be prepared by the reaction between:

A. bases and non-metals
B. metallic oxide and dilute acids
C. acidic oxide and water
D. active metal and water

Explanation

base + non-metal ⟶ no reaction

metallic oxide + dilute acid ⟶ salt + water

acidic oxide + water ⟶ Acid

active metal + water ⟶ base + hydrogen gas

Answer B

93. What would be the formula of the salt that might be produced when carbonic acid H_2CO_3 is combined with aluminum hydroxide $Al(OH)_3$?

A. $AlCO_3$
B. Al_4C_3
C. $Al_2(CO_3)_3$
D. Al_2CO_3

Explanation

$2Al(OH)_3 + 3H_2CO_3 \longrightarrow Al_2(CO_3)_3 + 6H_2O$

Answer C

94. Which one of the following acids can form three types of salts?

A. Carbonic acid
B. Perchloric acid
C. Phospheric acid
D. sulphuric acid

Explanation

➦ Triprotic acids form three types of salts

Na_3PO_4 – sodium phosphate

Na_2HPO_4 - sodium monohydrogen phosphate

NaH_2PO_4 – sodium dihydrogen phosphate

} Formed from H_3PO_4

95. Complete the following reactions and balance

a. $Fe + S \longrightarrow$

b. $CuO + H_2SO_4 \rightarrow$

c. $KHCO_3 + HNO_2 \longrightarrow$

d. $NaOH + HCN \longrightarrow$

e. $Zn + HI \longrightarrow$

f. $CaCl_2 + Na_2CO_3 \rightarrow$

Answer

a. $Fe + S \longrightarrow FeS$

b. $CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

c. $KHCO_3 + HNO_2 \longrightarrow KNO_2 + CO_2 + H_2O$

d. $NaOH + HCN \longrightarrow NaCN + H_2O$

e. $Zn + 2HI \longrightarrow ZnI_2 + H_2$

f. $CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

96. Write the formula and name of salts formed when KOH reacts with H_2SO_4

Explanation

$KOH + H_2SO_4 \longrightarrow KHSO_4 + H_2O$

Potassium hydrogen sulphate - $KHSO_4$

$2KOH + H_2SO_4 \longrightarrow K_2SO_4 + H_2O$

Potassium sulphate - K_2SO_4

97. How can you prepare sodium nitrate from sodium carbonate?

Explanation

When sodium carbonate reacts with nitric acid sodium nitrate, carbondioxide and water are formed.

$\Rightarrow Na_2CO_3 + 2HNO_3 \longrightarrow 2NaNO_3 + CO_2 + H_2O$

98. Name the acid and base pairs required to prepare

a. $Mg(NO_2)_2$ b. CuF_2 c. BaF_2 d. CaS

Answer

a. Magnesium hydroxide ($Mg(OH)_2$) and nitrous acid (HNO_2)
b. Copper (II) hydroxide ($Cu(OH)_2$) and hydrofluoric acid (HF)
c. Barium hydroxide ($Ba(OH)_2$) and hydrofluoric acid (HF)
d. Calcium hydroxide ($Ca(OH)_2$) and hydrosulphuric acid (H_2S)

99. List three methods used to make salts

Answer

- Direct combination of metals with non-metals
- Neutralization reaction
- Double displacement reaction

Some important salts

i Sodium chloride (NaCl)

It is commonly known as table salt.

- It is used for flavoring and preserving of foods
- It is used as a raw material to produce sodium, chlorine, sodium hydroxide, sodium hydrogen carbonate and sodium carbonate
- It is used in tannery to dry the skin and kill bacteria
- It has medical application as a component of ORS

ii Copper (II) sulphate ($CuSO_4$)

- It is useful in electroplating and to prevent fungal attack of leaves and vines as a mixture together with calcium hydroxide. A mixture of $CuSO_4$ and $Ca(OH)_2$ is called **Bordeausx**

iii Iron (III) chloride ($FeCl_3$)

- It is also called ferric chloride.
- It is used to treate waste water and feed additive.

iv Iron (II) sulphate ($FeSO_4$)

- It is used as iron tablets to protect patients from anaemia and wood preservative, water and sawage treatment; to manufacture ink

v **Barium sulphate ($BaSO_4$)**

- It is used as white pigment and taking x-ray photographs of the intestimal tract since it is opaque to x-rays.

vi **Hydrated calcium sulphate ($CaSO_4.2H_2O$)**

- It is also known as **gypsum.**
- It used to plaster walls and support fractured bones.

vii **Potassium nitrate (KNO_3)**

- It is commonly called **salt peter**
- It is used to make gun powder and fertilizer.

viii **Ammonium nitrate (NH_4NO_3)**

- It is mainly used as nitrogeneous fertilizer

ix **Aluminum sulphate($Al_2(SO_4)_3$)**

- It is mainly used in paper industry and water purification

x **Hydrated magnesium sulphate ($MgSO_4.7H_2O$)**

- It is also called Epsom salt
- It is used as laxative and for other medical purposes.

Illustrative examples

100. Which of the following salts is used to make fungicides?

A. $CuSO_4$ C. KNO_3

B. NaCl D. $BaSO_4$

Answer A

101. Identify the salt that is given as iron tablets to patients who suffer from anaemia among the following?

A. NaCl C. $FeSO_4$

B. NH_4NO_3 D. $FeCl_3$

Answer C

102. What are the components of Bordeausx?

Answer

$CuSO_4$, and $Cu(OH)_2$

103. Gun powder is a mixture. List its components

Answer:

KNO_3, C and S

Properties of salts

1. Solubility in water

⇒ Salts that contain the following are soluble in water

- ➦ Nitrate (NO_3^-)
- ➦ Ammonium (NH_4^+)
- ➦ Chlorate (ClO_3^-)
- ➦ Perchlorate (ClO_4^-)
- ➦ Alkali metals (Li, Na, K, Rb)
- ➦ Acetate (CH_3COO^-)

⇒ All halides except that of Ag^+, Hg^{2+}, and Pb^{2+} are soluble in water.

⇒ All sulphates except that of , Ba^{2+},Hg^{2+}, Pb^{2+} and Sr^{2+} are soluble in water.

⇒ Carbonates , phosphates and sulphides are insoluble in water except that of alkali metals and ammonium ion.

⇒ Sulphides of alkaline earth metals are soluble in water.

2. Some salts absorb water and other salts release water to the atmosphere

- ➦ Salts that absorb water from the atmosphere but remain solid are **called hydroscopic salts**
- ➦ Salts that absorb water from the atmosphere to form a solution are called **deliquescent salts**

Example

$CaCl_2$, $NaNO_3$, $FeCl_3$, $HgCl_2$, $LiNO_3$, $NaNO_3$

- ➦ Salts that lose their water of crystallization to the atmosphere are known as **efflorescent salts**

Example

$Na_2CO_3 . 10H_2O$, $Na_2SO_4 . 10H_2O$

3. Slats are electrolytes

- Salts transmit electricity in molten and aqueous state.

4. Salts are decomposed by heat

Thermal decomposition of carbonates

- Carbonates are salts of carbonic acid that decompose by heat to form metallic oxide and carbondioxide.

Example:

$CaCO_3 \rightarrow CaO + CO_2$

$Li_2CO_3 \rightarrow Li_2O + CO_2$

Carbonates of sodium and potassium Na_2CO_3 and K_2CO_3 are not decomposed by heat

ii Thermal decomposition of nitrates

- **Nitrates are salts of nitric acid. They give metallicoxide, nitrogen dioxide and oxygen upon decomposition.**

Example:

- $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
- $4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

Sodium nitrate and potassium nitrate give metal nitrites and oxygen upon decomposition by heat.

Illustrative Examples

104. All of the following salts are insoluble in water. EXCEPT

A. $BaSO_4$
B. Na_2SO_3
C. $CaCO_3$
D. AgCl

Explanation

Salts containing alkali metals are soluble in water.

Answer B

105. Which of the following salts cannot be decomposed by heat?

A. K_2CO_3

B. $KClO_3$

C. KNO_3

D. None

Answer A

106. Which of the following is deliquescent salt?

A. $Na_2SO_4 . 10H_2O$

B. $NaNO_3$

C. CuSO

D. $Na_2CO_3.10H_2O$

Answer B

107. Complete and balance the following

a. $NaNO_3 \xrightarrow{heat}$

b. $LiNO_3 \xrightarrow{heat}$

c. $Ca(NO_3)_2 \xrightarrow{heat}$

d. $MgCO_3 \xrightarrow{heat}$

Answer

a. $2NaNO_3 \xrightarrow{heat} 2NaNO_2 + O_2$

b. $4LiNO_3 \xrightarrow{heat} 2Li_2O + 4NO_2 + O_2$

c. $2Ca(NO_3)_2 \xrightarrow{heat} 2CaO + 4NO_2 + O_2$

d. $MgCO_3 \xrightarrow{heat} MgO + CO_2$

108. What types of salts releases a redish – brown gas nitrogen dioxide when they are decomposed by heat?

Answer: Nitrates

109. The process of absorbing water from the atmosphere by a solid to form a solution is called _____

Answer: deliquescence

7.4 Chemical tests of some ions in salts

1. **Flame test** is a qualitative test to identify metal ions in salts by using **nichome wire** (an alloy of nickel and chromium)

Example:

Metal ion in salt	Color of flame produced
Lithium (Li^+)	Crimson
Sodium (Na^+)	yellow
Potassium (K^+)	violet
Calcium (Ca^{2+})	Orange-red
Strontium (Sr^{2+})	crimson
Barium (Ba^{2+})	green

- Blue solution of copper indicates the presence of Cu^{2+} in the solution
- Pale – green precipitate formation when iron (II) compounds react with alkaline solution indicates presence of Fe^{2+}
- Pale – brown precipitate formation when iron compounds react with alkaline solution confirms the presence of Fe^{3+}

2. **Test of halides:** Mixing sodium halides with $AgNO_3$
   - When white precipitate is formed, confirms presence of Cl^- **ions.**
   - Formation of yellow precipitate indicates the presence of Br^- **ions.**
   - Formation of yellowish – green precipitates proves the presence of I^- **ions.**

3. **Test for sulphates (SO_4^{2-})**

When soluble salts of barium and sodium-sulphate are mixed, **white precipitate.** $BaSO_4$ is formed. This confirms the presence of sulphate ion.

4. **Test for nitrates (NO_3^-)**
   - Nitrates are identified by **Brown-ring test.** In a freshly prepared iron (II) sulphate if concentrated sulphuric acid is poured down the side of the test tube, the acid sinks to the bottom and a brown ring is formed this happens when there is nitrate ion in the solution.
   - The formula of brown ring is $FeSO_4.NO$.

Illustrative Examples

110. Which of the following group of salts give a white precipitate when a solution of barium chloride is added to them? (EGSECE.....2001E.C)

A. Nitrates
B. Chlorates
C. Permanganets
D. Sulphates

Answer D

111. Which of the following could form a black precipitate when lead acetate is added to it? (EGSECE.....1993E.C)

A. H_2SO_4
B. H_2S
C. SO_2
D. SO_3

Explanation

$Pb(CH_3COO)_2 + H_2S \longrightarrow PbS(s) + 2CH_3COOH$ Black precipitate-pbs

Answer B

112. The brown ring test is used to identify the presence of:

A. $N\bar{O}_3$
B. SO_4^{2-}
C. NO_2^-
D. SO_3^{2-}

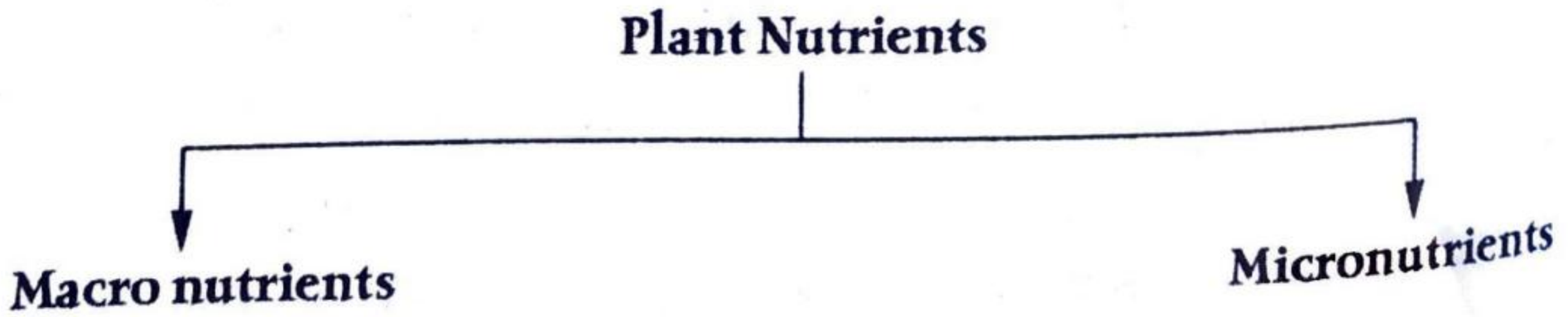
Answer: A

113. What is the laboratory apparatus used for flame tests?

Answer: Nichrome wire

7.5 Plant Nutrients

Plant nutrients are minerals required for the growth and development of plants. These nutrients are classified into two.



Required in large quantities	Required in small or trace amount
Example	**Example**
➦ Nitrogen	➦ Iron
➦ Phosphorus	➦ Cobalt
➦ Potassium	➦ Zinc
➦ Magnesium	➦ Manganese
➦ Calcium	➦ Molybdenum
➦ Sulphur	➦ Copper
➦ Carbon	➦ Boron
➦ Hydrogen	➦ Chlorine
➦ Oxygen	

The role of nitrogen, phosphrus and potassium in the plant growth

A. Nitrogen

- ➦ It is absorbed by plants in the form of nitrate ion (NO^-_3)
- ➦ It is used to make aminoacids, proteins, coenzymes and nucliec acid.
- ➦ It is also used in the synthesis of a green pigment chlorophyll.
- ➦ The leaves of plants turn yellow when they face lack of nitrogen.

B. Phosphorus.

- ➦ It is absorbed by plants in the form of $H_2PO^-_4$
- ➦ It facilitates early growth, root formation, quick maturity and promotes seed or fruit production.
- ➦ It has role in the formation of amino acids, protein, coenzymes, nucleic acid and high energy phosphate (ATP).

C. Potassium

- ➦ It is absorbed by plants in the form of potassium ion (K^+)
- ➦ It facilitates the process of photosynthesis
- ➦ It adjusts stomata movements

- ➦ It controls the turgidity of plants.
- ➦ It also increases solute potential.

Fertilizers

- ➦ Fertilizers are added to soil, to increase the growth, yield and nutrient value of crop. They are also used to replace the nutrients lost from the soil. There are two types of fertilizers. These are:

i **Organic /natural/ fertilizers**

- ➦ They are obtained from animals and plants.

Example:

animal dung, urine and substance from the decay of plants

ii **Synthetic or artificial fertilizers**

- ➦ Contain all the three major elements nitrogen, phosphorus and potassium (NPK)

Example:

Diammoniummonohydrogen phosphate ($(NH_4)_2HPO_4$), DAP, ammonium sulphate ($(NH_4)_2SO_4$), potassiumnitrate(KNO_3)

$$2NH_3 + H_3PO_4 \longrightarrow (NH_4)_2\ HPO_4$$

DAP

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

Pesticides

- ➦ These are chemicals that kill pests which affect the growth of plants, most of pesticides are organic compounds.

Examples:

Fumigants – HCN, SO_2, CS_2 and PH_3

Inorganic pesticides – $CuCl_2$, $Cu(OH)_2$, $Cu(NO_3)_2$, $CuSO_4$, Cu_2O and $CuCO_3$

- $CuSO_4$ and Cu_2O are fungicides
- HCN, SO_2, PH_3, $CuCl_2$, $Cu(OH)_2$, $Cu(NO_3)_2$, and $CuCO_3$ are used as insecticides.

Herbicides/Weed killers/

- These are chemicals that remove weeds.
- Weeds are unwanted plants

Example

Borax – $Na_2B_4O_7.10H_2O$

Sodiumchlorate – $NaClO_3$

Hydrated iron (II) sulphate – $FeSO_4.7H_2O$

Ammonium sulphanate – $(NH_4)SO_3 – NH_2$

Illustrative examples

114. Which of the following set of elements are macronutrietns of plants? (EGSECEC……2008E.C)

A. Zn, Fe, B
B. Cl, Cu, Mo
C. Fe, Clo, Mn
D. P, K, N

Answer D

115. Which of the following is a pesticide?

A. Borax
B. Copper (II) sulphate
C. potassium nitrate
D. Sodium chlorate

Explanation

- Sodium chlorate and borax are herbicides
- Potassium nitrate is a fertilizer

Answer:B

116. What type of fertilizer is a complete fertilizer?

Answer:

A fertilizer that contains the three major elements potassium, phosphorus and nitrogen.

Solved problems

117. Two solution A and B have pH values of 3 and 6. How many times greater is the hydrogen ion concentration in Solution A than that of Solution B?

Solution:

$$\left[H^+\right]_A = 10^{-pHA} \qquad pH_A = 3$$

$$\left[H^+\right]_B = 10^{pH_B} \qquad pH_B = 6$$

$$\frac{\left[H^+\right]}{\left[H^+\right]_B} = \frac{10^{-pH_A}}{10^{-pH_B}}$$

$$\frac{\left[H^+\right]_A}{\left[H^+\right]_B} = \frac{10^{-3}}{10^{6}}$$

$$\frac{\left[H^+\right]_A}{\left[H^+\right]_B} = 10^{-3+6}.$$

$$\left[H^+\right]_A = 1000\left[H^+\right]B$$

$\therefore$ 1000 times greater

The lower the value of pH, the higher is the concentration of hydrogen ion.

118. Three different acids, HX, HY and HZ have pH value 1, 4 and 3 respectively. Arrange these acids in increasing order of acid strength.

Solution:

HY < HZ < HX, because the lower the pH value, the stronger is the acid.

119. Calculate the pH of 1.0×10^{-3} $Ba(OH)_2$ solution assuming complete ionization.

Solution

$$Ba(OH)_2 \longrightarrow 2OH^- + Ba^{2+}$$

$$1M = 2M$$

$$1 \times 10^{-3}M = \quad ?$$

$$\left[OH^-\right] = \frac{1\times10^{-3}M\times2M}{1M} = 2\times10^{-3}M$$

$$pH = 14 + \log[OH^-]$$

$$= 14 + \log^{2\times10^{-5}}$$

$$pH = 11.30$$

120. Determine the concentration of hydroxide ion in H_2SO_4. Solutions assume complete ionization.

Solution:

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$$[H^+] = \frac{2\times10^{-5}\times2M}{1M} = 4\times10^{-5}M$$

$$[OH^-] = \frac{K_w}{[H^+]} = \frac{10^{-14}M^2}{4\times10^{-5}M}$$

$$[OH^-] = 2.5\times10^{-10}M$$

121. When sodium hydroxide is added to dilute hydrochloric acid solution in a beaker, what happens to the pH, $[H^+]$ and $[OH^-]$ values.

Solution:

- The pH of the solution in the beaker increases
- The pOH of the solution decreases
- The $[H^+]$ decreases
- The $[OH^-]$ increase

122. Calculate the pH of a solution formed by dissolving 2.8g of KOH in 100mL of solution

Solution:

First determine the concentration of the base m=2.8g

$$M_{KOH} = 39 + 16 + 1 = 56g/mol$$

$$M = \frac{m}{M_{KOH}\times V} = \frac{2.8g}{56g/mol\times0.1L} = 0.5mol/L$$

V = 100mL = 0.1L

$[KOH]=[OH^-]=0.5mol/L$

$pH = 14 + \log^{[OH]}$

$pH = 14 + \log^{0.5}$

$pH = 13.70$

123. State the name of a suitable acid and alkali you can use to make each of the following salts.

a. Sodium sulphate

b. Potassium phosphate

c. Ammonium nitrate

Solution:

a. Sulfuric acid and Sodium hydroxide

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2$

Sodium sulphate - Na_2SO_4

b. Phosphoric acid and potassium hydroxide

$H_3PO_4 + KOH \longrightarrow K_3PO_4 + 3H_2O$

Potassium phosphate - K_3PO_4

c. Nitric acid and ammonia

$HNO_3 + NH_3 \longrightarrow NH_4NO_3$

Ammonium nitrate - NH_4NO_3

124. Arrange the following acids in decreasing order of acid strength $HClO$, $HClO_2$, $HClO_3$ and $HClO_4$

Solution:

$HClO_4 > HClO_3 > HClO_2 > HClO$ because as number of oxygen atom increases, strength of acidity also increases.

$HClO_4$ – perchloric acid

$HClO_3$ – chloric acid

$HClO_2$ – Chlorous acid

$HClO$ – Hypochlorous acid

125. Find the conjugate acid of

a. HS^-

b. PO_4^{3-}

c. KOH

Solution:

Base + $H^+ \longrightarrow$ Conjugate acid

a. $HS^- + H^+ \longrightarrow H_2S$

b. $PO_4^{3-} + H^+ \rightarrow HPO_4^{2-}$

c. Has no conjugate acid, because it is not Bronsted – Lowery base

126. Identify the conjugate base of

a. N_2H_5 b. H_2SO_4 c. H_2PO_4 d. CN^-

Solution:

Acid – $H^+ \longrightarrow$ Conjugate base

a. $N_2H_5^+ - H^+N_2H_4$

b. $H_2SO_4 - H^+ \longrightarrow HSO_4^{2-}$

c. $H_2PO_4^- - H^+ \rightarrow HPO_4^{2-}$

d. Has no conjugate base since it is not Bronsted – lowry acid.

127. Write down the formula for the precipitate which is produced when pairs of aqueous solution are mixed as follows:

a. NH_4Br and $AgNO_3$

b. $CaCl_2$ and K_2CO_3

c. $Ba(NO_3)_2$ and Na_2SO_4

Solution:

Precipitate means an insoluble salt in water

a. $NH_4Br(aq) + AgNO_3(aq) \longrightarrow AgBr(s) + NH_4NO_3(aq)$

precipitate - AgBr

b. $CaCl_2(aq) + K_2CO_3(aq) \longrightarrow CaCO_3(s) + 2KCl(aq)$

precipitate - $CaCO_3$

c. $Ba(NO_3)_{2(aq)} + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2KCl(aq)$

precipitate - $BaSO_4$

128. The pH of 250ml sulfuric acid solution 1. Calculate the mass of sulfuric acid required to prepare this solution assume complete ionization of the acid.

Solution:

V = 250ml = 0.250l

pH = 1

$[H^+] = 10^{-1}M = 0.1M$

$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

1M = 2M

? = 0.1M

$$M = [H_2SO_4] = \frac{1M \times 0.1M}{2M} = 0.05M$$

$$M_w = 2\times1 + 32\times1 + 4\times16 = 98g/mol$$

$$M = \frac{m_{H_2SO_4}}{M_{H_2SO_4} \times V}$$

$$m_{H_2SO_4} = 0.05mol/L \times 98g/mol \times 0.25L$$

$$m_{H_2SO_4} = 1.225g$$

129. State some laboratory techniques that help to discover the acidity or basicity of a solution?

Solution:

i Determination o f the pH of the solution

ii Determination of hydrogen or hydroxide ion concentration

iii Testing the solutions with appropriate indicators.

130. How many militers of 18M sulfuric acid is required to make a 500mL of 0.36M H_2SO_4 solution?

Solution:

During dilution number of mole of sulfuric acid remain same but molarity and volume change.

$$M = \frac{n}{V} \qquad n = M.V$$

$n_1 = M_1.V_1$

$n_2 = M_2.V_2$

but $n_1 = n_2$ (unchanged)

$M_1V_1 = M_2V_2$

This implies ,

M_1 – initial molarity M_2 – final molarity

V_1 – initial volume V_2 – fi1nal volume

$M_1 = 18M$ $M_2 = 0.36M$

$V_1 = ?$ $V_2 = 500mL$

$M_1V_1 = M_2V_2$

$V_1 = 10mL$

V_{H_2O} added $= V_2 - V_1 = 500mL - 10L = 490mL$

131. An unknown substance reacts with marble chips releasing a gas that extinguishes a burning splint and neutralizes a basic solution. The unknown substance could most likely be _____.

Solution:

A gas that extinguishes fire is carbon dioxide. An unknown substance that release carbon dioxide when it reacts with marble chips is an acid.

Example

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

Marble chips

132. What type of salt forms a white precipitate when treated with a solution of soluble salts of barium?

Solution:

Formation of white precipitate when soluble salt of barium added to a given solution indicates the presence of sulphate ion.

Example

$Na_2SO_4(aq) + Ba(NO_3)_2(aq) \longrightarrow BaSO_4(s) + 2NaNO_3(aq)$

133. What is the molarity of a solution obtained by dissolving 0.01 mol of NaCl in 500 mL of solution?

Solution

$M = \frac{n}{v}$ n= 0.01 mol, V=500mL = 0.50L

$M = \frac{0.01mol}{0.50L}$

M=0.02mol/L = 0.02M

134. Arrange the following with increasing order of concentration of hydroxide ion.

a. pH = 4.26
b. pH = 7.74
c. pH = 9.82
d. pH = 12.60

Solution:

The concentration of hydroxide ion is directly related with the pH value of a given solution.

$pH = 14 + \log [OH^-]$

$\Rightarrow a < b < c < d$

135. What is the quantity of water, in mL, required to prepare 0.50M of HCl from a concentrated solution of 3.5M in 50ml is?

Solution:

During dilution, number of mole of solute remains same while to concentration and volume of the solution change.

$\Rightarrow M_1V_1 = M_2V_2$

$M_1 = 3.5M$

$V_1 = 50mL$

$M_2 = 0.5M$

$V_2 = ?$

$$V_2 = \frac{M_1V_1}{M_2} = \frac{3.5M \times mL}{0.5M} = 350mL$$

$$V_{H_2O} = 350 - 50 = 300 \text{ ml}$$

136. How many mL conc. HNO_3 from a conc. $13MNO_3$ is required to prepare 500mL of 0.1M HNO_3?

Solution

$$V_1 = \frac{M_2V_2}{M_1} = \frac{0.1M \times 500mL}{13M} = 3.85mL$$

$$V_{H_2O} = 500 - 3.85 = 396.15mL$$

Hence, 3.85mL of conc, HNO_3 and 496.15mL of water are required.

137. List some acid – base indicators.

Solution:

Indicator is a substance used to show the presence of acid or base.

Indicator	Color of indicator in	
	Acidic solution	Basic solution
Litumus	Red	Blue
Phenol phthalien	Color less	Pink
Methyl orange	Red	Yellow
Thymol blue	Red	Yellow
Bromocreso green	Yellow	Blue
Bromothymol blue	Yellow	Blue

Universal indicator is a mixture of indicators often sold readymade as solution which can indicate pH values usually over a range of about 3-11 by successive changes of color.

138. It is required to prepare one liter of 2M solution of NaCl. How many grams of NaCl are required? (Atomic mass: Na=23, Cl=35.5)

Solution:

$V=1L$, $M_{NaCl} = 23 + 35.5 = 58.5g/mol$

$M = 2M$

n=?

n=M×V

$$n = 2\frac{mol}{L} \times 1L = 2mol$$

$$m_{NaCl} = n_{NaCl} \times M_{NaCl}$$

$$m_{NaCl} = 2mol \times 58.6\frac{g}{mol} = 117g$$

Hence, 117g of NaCl is required to prepare 1L of 2M NaCl solution.

139. What volume of 0.5M HCl contain 10 mol of HCl?

Solution:

M = 0.5M

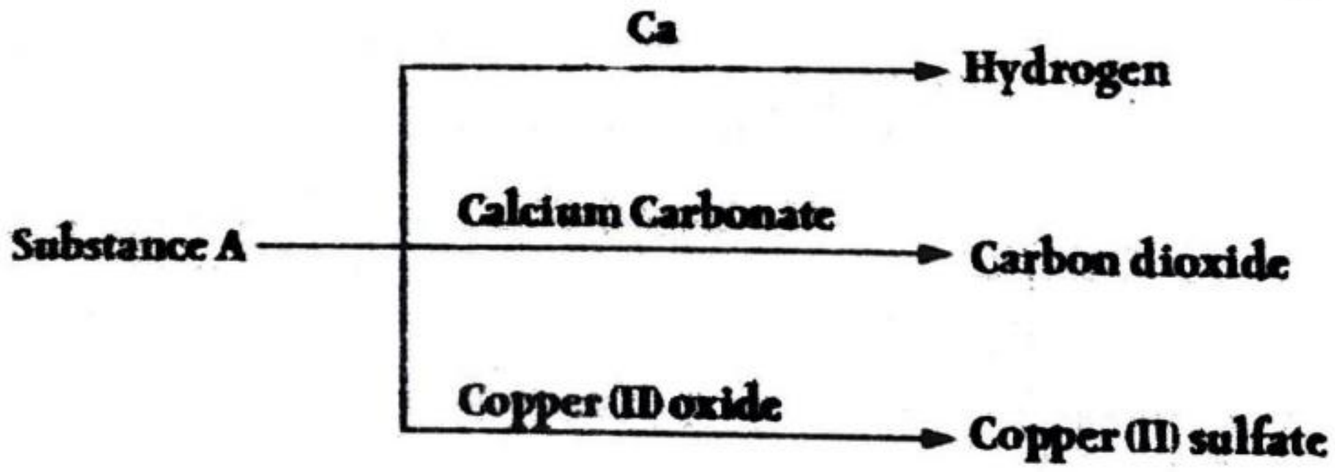
n=10mol

v=?

$$M = \frac{n}{V}$$

$$V = \frac{10mol}{0.5mol/L} = 20L$$

140. Some reactions of a substance, A, are shown in the diagram below.



What type of substance is A?

Solution:

A substance that reacts with metal to form hydrogen gas, with carbonates to form carbon dioxide and with metal oxide, salt is acid. Therefore, substance A should be an acid.

141. The pH of a solution of HCl in water is found to be 2. What volume of water would you add to 1.00L of this solution to raise the pH to 3.

Solution:

For strong monoprotic acid, concentration of hydrogen ion is equal to concentration of acid.

$\Rightarrow [H^+] = [Acid]$

$[H^+]_1 = [HCl]_1 = 10^{-2} = 0.01M = M_1$

$[H^+]_2 = [HCl]_2 = 10^{-3} = 0.001M = M_2$

$V_1 = 1.00L$

$M_1V_1 = M_2V_2$

$$V_2 = \frac{0.01M \times 1L}{0.001M}$$

$V_2 = 10L$

$V_{water\ added} = V_2 - V_1$

$V_{water\ added} = 10L - 1L$

$V_{water\ added} = 9L$

142. Identify Lewis acids and bases in these reactions.

a. $Cr^{2+} + 6H_2O \longrightarrow [Cr(H_2O)_6]^{3+}$

b. $Al(OH)_3 + OH^- \longrightarrow [Al(OH)_4]^-$

c. $SnCl_4 + 2Cl^- \longrightarrow [SnCl_6]^{2-}$

Solution:

	Lewis acid	Lewis base
a.	Cr^{3+}	H_2O
b.	$Al(OH)_3$	OH^-
c.	$SnCl_4$	Cl^-

143. Write the three most important acids and bases.

Solution:

i Sulphuric acid (H_2SO_4)

ii Hydrochloric acid (HCl)

iii Nitric acid (HNO_3)

iv Sodium hydroxide (NaOH)

v Potassium hydroxide (KOH)

vi Calcium hydroxide ($Ca(OH)_2$)

144. Classify the following compounds as oxide, acid, base or salt,

a. Li_2O

b. HCN

c. $Fe(OH)_3$

d. H_3PO_2

e. $ZnCl_2$

f. Cr_2O_3

g. $Cu(OH)_2$

h. KNO_3

Solution:

a and f are oxides

b and d are acids

c and g are bases

e and h are slats

145. Classify the following properties of acid, base or both.

a. Have sour taste

b. Have bitter taste

c. Change color of indicators

d. React with active metals

e. Are electrolytes

f. pH < 7

g. pOH< 7

h. release hydroxide ion

i. release hydronium ion

Solution

a. acid

b. base

c. both acid and base

d. acid

e. both acid and base

f. acid

g. base

h. base

i. acid

146. Classify the following salts as soluble or insoluble salt in water.

a. NH_4Cl
b. KCN
c. AgBr
d. $CaCO_3$
e. $PbSO_4$
f. $MgCl_2$
g. FeS
h. $Fe_3(PO_4)_2$
i. $Cu(NO_3)_2$
j. $Zn(ClO_4)_2$

Solution:

a, b, f, i and j are soluble salts

c, d, e, g and h are insoluble salts

Supplementary Exercise on Unit Seven

I. Choose the correct answer from the given alternatives.

1. What is the concentration of nitric acid (HNO_3) solution with pH of 2?

A. 2M B. 0.2M C. 0.01M D. 0.1M

2. According to the Bronsted – Lowry concept, which of the following cannot function as an acid?

A. HSO_4^-
B. HNO_3
C. H_2CO_3
D. PCl_5

3. Which of the following is weak base?

A. HSO_4^-
B. $Fe(OH)_3$
C. KOH
D. $Ba(OH)_2$

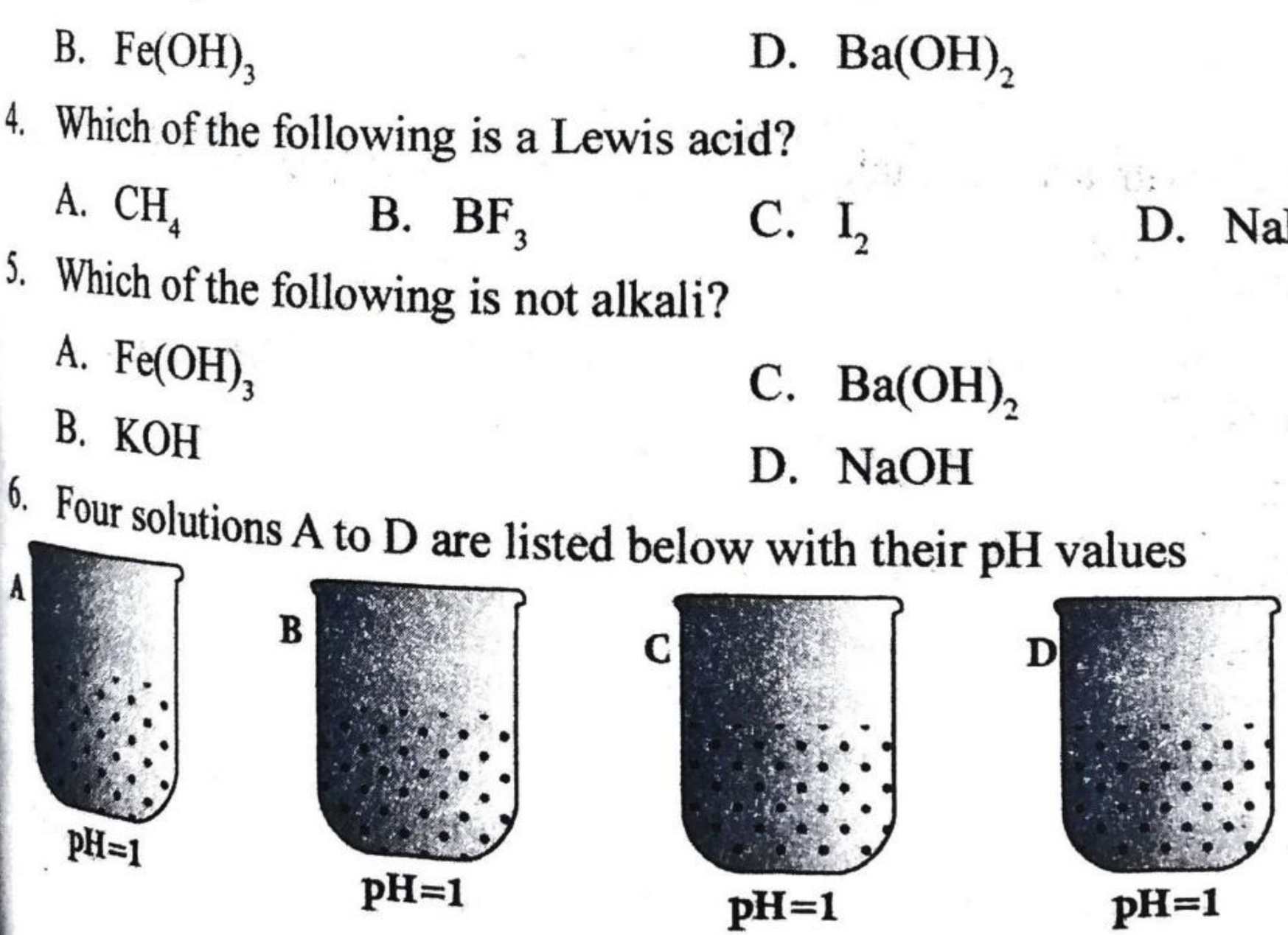
4. Which of the following is a Lewis acid?

A. CH_4 B. BF_3 C. I_2 D. NaH

5. Which of the following is not alkali?

A. $Fe(OH)_3$
B. KOH
C. $Ba(OH)_2$
D. NaOH

6. Four solutions A to D are listed below with their pH values



Which one is binary salt?

A. Solution A
B. Solution B
C. Solution C
D. Solution D

7. Which one is binary salt?

A. $NaNO_3$
B. KCl
C. $CaCO_3$
D. KCN

8. Which of the following is not true about a 0,1M strong acid HA?

A. pH = 1
B. $[H^+] = [HA]$
C. $[H^+]=[A^-]$
D. $[H^+]>[HA]$

9. Which of the following represents a neutral solution?

A. $[H^+]=10^{-5}M$
B. $[OH^-] = 2.6 \times 10^{-7}M$
C. $[OH^-] = 10^{-7}M$
D. $[H^+] = 2 \times 10^{-8}M$

10. Which of the following commonly known as slacked lime?

A. CaO
B. $Ca(OH)_2$
C. KOH
D. None

11. Which of the following does not contain metal ion?

A. NH_4OH
B. $Ba(OH)_2$
C. $Ca(OH)_2$
D. KOH

12. If a strong base is splashed into the eye of a person, the compound that should be used as first aid treatment is

A. Sulfuric acid
B. Sodium carbonate
C. Acetic acid
D. Magnesium hydroxide

13. Which of the following is not Arrhenius acid?

A. HF
B. HCN
C. BCl_3
D. HNO_3

14. The color of a solution X, turns to pink when a drop of phenophythalien is added to it, solution x should be:

A. Acidic
B. basic
C. Neutral
D. None

15. Which of the following is not ternary salt?

A. NaCl
B. KCO_3
C. KCN
D. $Al_2(SO_4)_3$

16. Which of the following is polyprotic acid?

A. HCOOH
B. HCN
C. $H_2C_2O_2$
D. HNO_3

17. All of the following are basic oxides except

A. Fe_2O_3 B. Al_2O_3 C. Na_2O D. CaO

18. A bottle contains a solution that is labeled pOH = 1. The solution of substance in the bottle can be described as

A. Weakly acidic
B. Strongly acidic
C. weakly basic
D. strongly basic

19. Which one of the following statements is correct about the properties of acids?

A. The pH of an aqueous solution of an acid increases as more water is added to it.
B. The pH of an acid increases with an increase in the concentration of the acid.
C. Acids change the color of methyl red to yellow.
D. Acids are electron pair donors

20. The Brown ring test is used to identify

A. Nitrates
B. Nitrities
C. sulphates
D. Sulphites

21. Which of the following is true about oxides?

A. All non-metallic oxides are acidic oxides
B. All metallic oxides are basic oxides
C. The oxidation state of oxygen in peroxides is -2.
D. Neutral oxides are very few in number when compared to other oxides.

22. Hardness causing minerals can be precipitated from water with

A. NaCl
C. Na_2CO_3
B. B. $BaSO_4$
D. $CaCl_2$

23. Element X reacts with chlorine to from a compound with the formula XCl_2. The oxide of this element is basic. Element X is

A. Na
C. P
B. Ca
D. C

24. The concentration of H^+ in a solution is 2×10^{-4} M. The OH^- concentration is

A. 2×10^{-4}M
C. 2×10^{-10}M
B. 5×10^{-11}M
D. 5×10^{-8}M

25. Which of the following oxides are acidic?

i	CaO	iii	NO_2	v	SO_3
ii	Na_2O	iv	CO	vi	P_2O_5

A. III, V and VI
C. I and II
B. III, IV, V and VI
D. I, II and V

26. Which of the following substances is both a strong acid and strong oxidizing agent?

A. HCl
C. H_2CO_3
B. H_2SO_4
D. H_3PO_4

II. Short Answer

27. What volume of water is required to 50mL of a solution of 0.2M HCl to produce 0.01M HCl solution?

28. Classify the following substances as macronutrient or micronutrient.

a. Potassium
b. Iron

c. Nitrogen

d. Cobalt

e. Carbon

f. Phosphorus

29. Name the following compounds

a. $Ca_3(PO_4)_2$

b. Cu_2O

c. $NaNO_2$

d. $Fe_2(CO_3)_3$

e. MnO_2

f. K_2SO_3

g. MgN_2

30. Complete the following reactions

a. Acid + Active metal $\longrightarrow$

b. Acid + Carbonatte $\longrightarrow$

c. Sulphite + Acid $\longrightarrow$

d. Base + Acid $\longrightarrow$

31. What is pH scale? On the pH scale, which number represents the strongest acid and which number represent the strongest bases?

32. Define and give examples

a. Oral rehydration salt (ORS)

b. Deliquescent salt

c. Hygroscopic salt

d. Effloresceent salt

33. Write one application of flame colors.

34. Write the different types of fertilizers.

35. Write the balanced chemical equation for the thermal decomposition of potassium nitrate (KNO_3).

36. Identify the Lewis base in the following reactions.

a. $Ni + 4CO \longrightarrow Ni(CO)_4$

b. $Fe^{3+} + 6NH_3 \longrightarrow [Fe(NH_3)_6]^{3+}$

c. $O^{2-} + CO_2 \longrightarrow CO_3^{2-}$

UNIT 8

Electrochemistry

Unit Outline:

8.1 Electrical Conductivity

8.2 Electrolytic cell

8.3 Galvanic Cell

Introduction

➦ Electrochemistry constitutes one of the most important interfaces between chemistry and everyday life. Every time you start your car, turn on your calculator, look at your digital watch or listen to radio at the beach, you are depending on electro chemical reactions. Batteries are the result of electrochemistry.

Electrochemistry is defined as the study of the intercharge of chemical and electrical energy. It is primarily concerned with two processes that involve oxidation – reduction reactions: the generation of an electric current from a chemical reaction and the use of a current to produce chemical change.

➦ The apparatus where energy conversion takes place is called **electrochemical cell.**

Electrochemical cell

Electrolytic cell
Electrical energy ⟶ chemical energy

Galvanic cell
Chemical energy ⟶ Electrical energy

8.1 Electrical conductivity

➦ It is the ability of substance to transmit electricity. The substances that transmit electricity are called **electrical conductors.**

➦ There are two types of conduction. These are metallic and electrolytic conductions.

Table 8.1. The difference between electrolytic and metallic conductions

Metallic conductions	Electrolytic conductions
➦ are also called electronic conductions because the charge carriers are free electrons.	➦ are also called ionic conductions because the charge carriers are ions
➦ transmit electricity without undergoining chemical changes	➦ transmit electricity by undergoing chemical changes
➦ Conduct electricity in all states	➦ conduct electricity in aqueous or molten states only.
Example: Metals	Example: salts, acids and bases

Strong electrolytes – dissociate almost completely in water

- produce large amount of ions that carry charge

Example:

- strong acid
- Strong base
- Molten salts

Weak electrolytes: ionize very slightly in water

- produce small amounts of ions when dissolved in water

Example:

- weak acids
- Partially soluble salts
- Weak bases

Non-electrolytes – are solutions that donot transmit electricity due to the absence of charge carriers.

Example:

- Sugar solution
- ethanol solution
- non-polar compounds in non-polar solvents

Illustrative examples

1. Which of the following is weak electrolytes? A solution of (EGSECE…..2008E.C)
   A. Sulphuric acid
   B. Acetic acid
   C. Nitric acid
   D. hydrochloric acid

Explanation: Sulphuric acid, nitric acid and hydrochloric acid are strong acids. However, acetic acid is a weak acid and weak electrolyte.

Answer B

2. The capacity of a substance to transmit electricity is termed as
   A. Conductivity
   B. Fluidity
   C. resistivity
   D. permittivity

Answer A

3. What are weak electrolytes?

A. They are ionic and polar covalent compounds

B. They release limited number of ions in water

C. They are strong acids and bases

D. They include sugar, alcohols and hydrocarbons

Answer B

4. Of the following substances one is an electrical conductor in solid state. Which one is it?

A. Sodium chloride

B. Sulphur crystal

C. potassiumhydroxide

D. Graphite

Explanation

⇒ Salts and bases in solid state do not transmit electricity because the ions are not free to carry charge. Non-metals except graphite are non-conductors.

Answer D

5. Which of the following are major charge carriers in electrolytic conduction?

A. Water molecules

B. Freely moving electrons

C. Cations and anions

D. neutral atoms

Answer C

6. Which of the following conduct electricity without undergoing any chemical change?

A. Sulphuric acid solution

B. Sodium chloride solution

C. Copper metal

D. Molten lead bromide

Explanation: Metals transmit electricity without undergoing chemical change.

Answer C

7. Which of the following does not transmit electricity?

A. KCl(1)

B. $CaCl_2(aq)$

C. Aluminium

D. $MgF_2(s)$

Explanation

⇒ l-molten, aq-aqueous solution and s-solid. Ionic compounds in solid state do not transmit electicity.

Answer D

8. Classify the following as strong electrolyte, weak electrolyte or non-electrolyte.
   a. Ammonia solution
   b. Ethanol solution
   c. Sodium nitrate solution
   d. Hydrofluoric acid solution
   e. Molten potassium chloride
   f. Sodium hydroxide solution
   g. Solid calcium chloride

Answer

a and d are weak electrolyte

b and g are non-electrolyte

c, e and f are strong electrolytes

9. Classify the following as metallic or electrolytic conductors.
   a. Gold
   b. Fused $CaCl_2$
   c. aluminum
   d. KCl (aq)

Answer

a and c are metallic conductors

b and d are electrolytic conductors

10. What are the two electrochemical cells?

Answer

Electrolytic and Galvanic cells

Conducting apparatus

➦ It is the instrument used to test the conductivity of substances.

➦ Its components are:
   - Connecting wire
   - Electrodes
   - Direct current source (battery)
   - Switch
   - Light bulb
   - Beaker

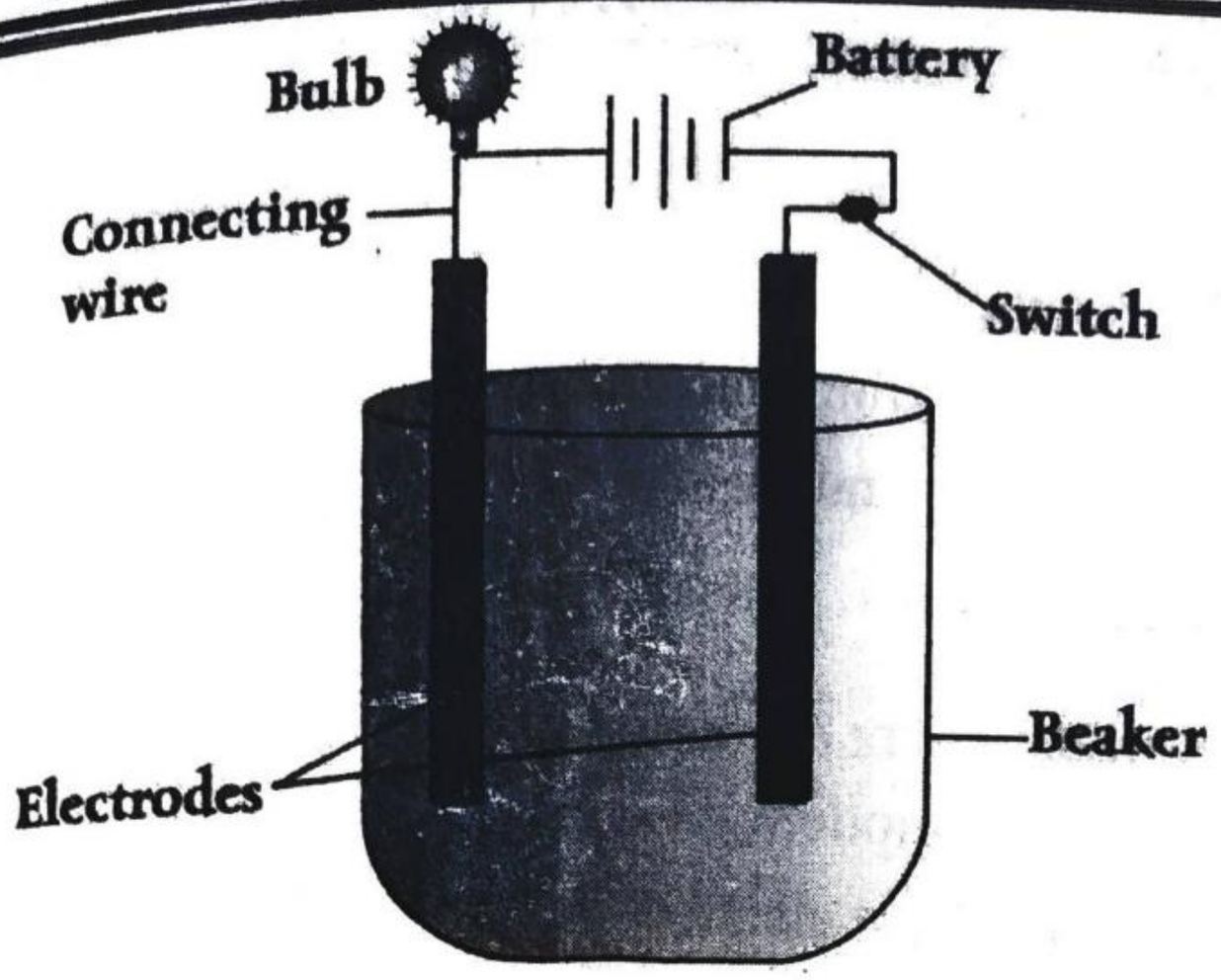


Figure 8.1 Conducting apparatus

8.2 Electrolytic cell

- It is an electrochemical cell which converts electrical energy into chemical energy.
- Electricity causes non-spontaneous reaction to occur.
- Its major components are:
  - Connecting wire
  - Electrodes
  - Direct current source
  - Electrolyte

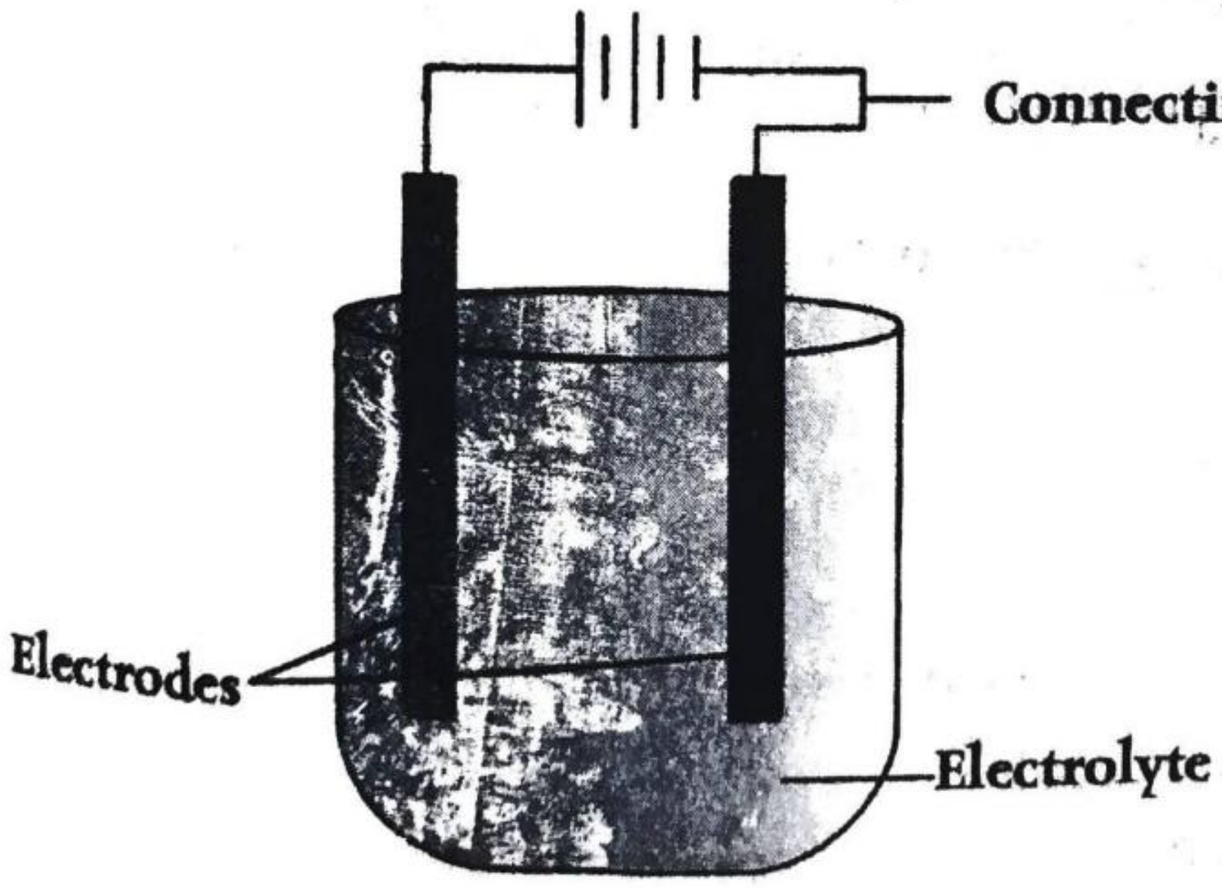


Figure 8.2 Electrolytic cell

Electrodes

- They are conducting roads
- They conduct electrons into and out of an electrolyte
- They are often a metal plate or wire or a piece of graphite

Classification of electrodes

1. **Anode** is electrode where oxidation half reaction occurs. Use AnOx to rember it easily. (An-anode, Ox- oxidation)
   - It is electrode where electrons leave the electrolyte
2. **Cathode** is electrode where reduction half reaction takes place. Use **'car'** to remember it easily. (ca-cathode, r-reduction)

Based on their nature, electrodes are classified

i **Active electrode**

- It reacts with electrolyte or product of electrolysis.
- It affects the result of electrolysis.

Example:

Zinc, magnesium, aluminum, copper etc

i **Inert electrode**

- It does not react with electrolyte or product of electrolysis.
- It does not affect the result of electrolysis.

Example:

graphite, gold, platinum etc

Electrolysis

- It is the process of converting electrical energy into chemical energy.
- It is the decomposition caused by electricity
- It is redox process.
- It takes place in electrolytic cell.

During electrolysis:

- The electrolyte decompose, so electrical energy causes a chemical change.
- Anode is positive and cathode is negative.
- Cations migrate toward cathode
- Anions move toward anode
- Electrons flow from anode to cathode in the external circuit.
- Battery is a driving force for electrons to move from anode to cathode.
- Electrons are released at anode and consumed at cathode.
- Electrons lost at anode are equal to electrons gained at cathode

Example

Electrolysis of molten or fused lead (II) bromide, $PbBr_2$

$$PbBr_2 \xrightarrow{melting} Pb^{2+} + 2Br^-$$

- Pb^{2+} moves toward cathode and discharged there.
- Br^- migrates toward anode and discharged there.

Chathod reaction: $Pb^{2+} + 2e^- \rightarrow Pb$ .....reduction

Anode reaction: $2Br^- \longrightarrow Br_2 + 2e$ ..... Oxidation

Cell reaction: $Pb^{2+} + 2Br^- \xrightarrow{electrolysis} Pb + Br_2$..... Redox reaction

Illustrative Examples

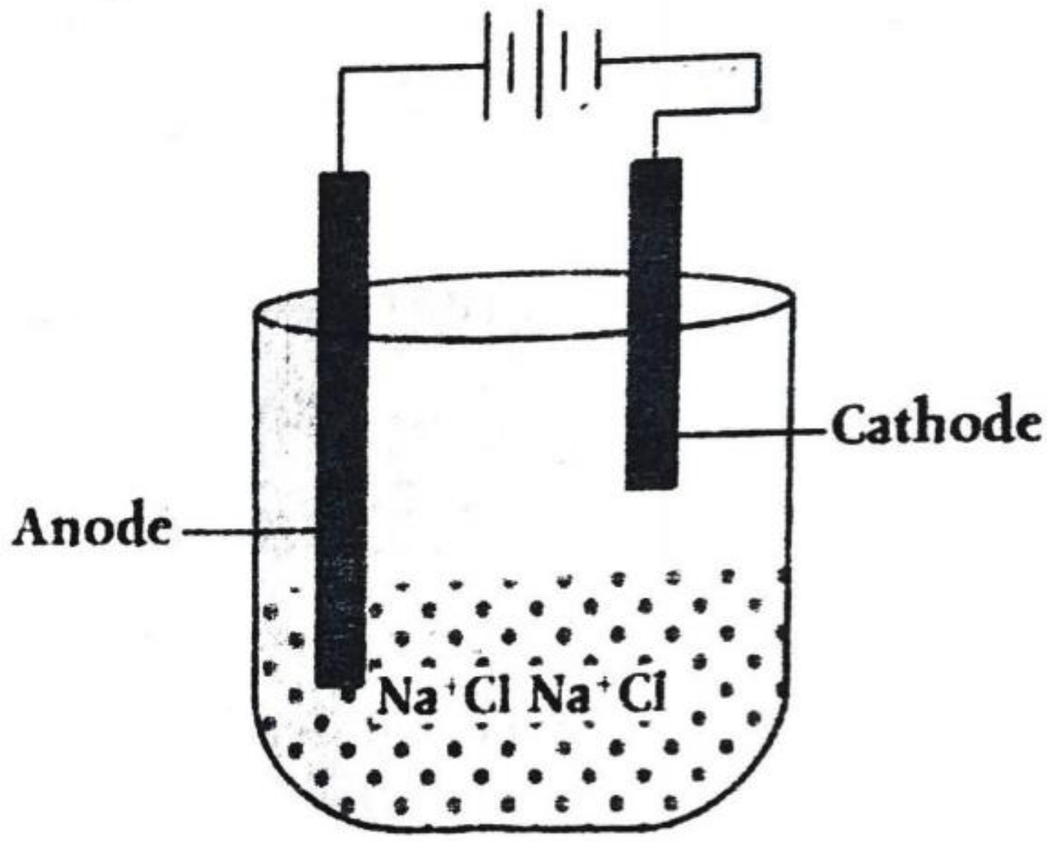
11. Which of the following combinations is correctly matches? (EGSECE..... 2008E.C)
    A. Cation $\longrightarrow$ an ion that loses electron/s/ to form atoms
    B. Anode $\longrightarrow$ a positive electrode in an electrolytic cell
    C. Cathode $\longrightarrow$ an electrode that attrats anions
    D. Anion $\longrightarrow$ an ion that gains electron/s/ to form atoms.

Explanation

- ➦ Cation is positive ion that gains electron/s/ to form atoms.
- ➦ Anion is negative ion that loses electron/s/ to form atoms.
- ➦ Anode attracts anions and cathode attracts cations.

Answer B

12. The following diagram shows the electron/s of molten sodium chloride using an electrolytic cell.



Which statement is true about this cell?

A. Electrons flow from cathode to anode
B. Reduction takes place at the positive electrode.
C. Chloride ion is discharged at the anode
D. Sodium ion is discharged at the anode

Explanation

- ➦ Electrons flow from anode to cathode
- ➦ Reduction takes place at negative electrode
- ➦ Sodium ion is discharged at cathode

Answer C

13. Which of the following represents the reduction half-reaction for the electrolysis of molten silver bromide? (EGSECE……..2008E.C)

A. $2Br^- \longrightarrow Br_2 + 2e$
B. $Br_2 + 2e^- \longrightarrow 2Br^-$
C. $Ag^+ + 1e^- \longrightarrow Ag$
D. $Ag \longrightarrow Ag^+ + 1e^-$

Explanation

$Ag^+ + 1e^- \longrightarrow Ag$ – reduction half – reaction

$2Br^- \longrightarrow Br_2 + 2e$ – oxidation half – reaction

Answer C

14. During electrolysis process

A. Cations lose electrons to form atoms

B. Cations are oxidized

C. Anions are atrtracted to the positive electrodes

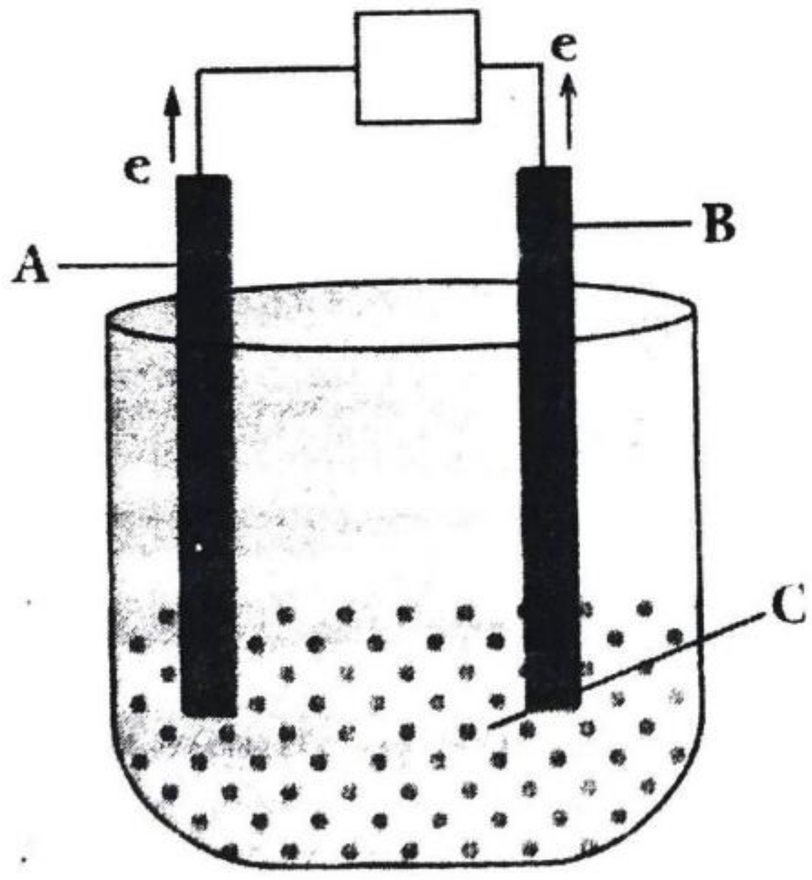
D. Anions are reduced

Explanation

- Cations gain and anions lose electrons
- Cations are reduced and anions are oxidized

Answer C

15. The diagram below shows electrolysis of molten $PbCl_2$ using chemically inert electrodes



Which of the following is correct?

A. Electrons flow from cathode to anode

B. Lead ion acts as a reducing agents

C. Chloride ion is reduced

D. Lead ion is discharged at the cathode

Explanation

- ➦ Electrons flow from anode to cathode
- ➦ Cations are oxidizing agent and anions are reducing agent
- ➦ Lead ion is reduced and chloride ion is oxidized.

Answer D

16. Which statements describes where oxidation and reduction half-reaction occur in an operating electrochemical cell?

A. Oxidation and reduction both occur at the anode

B. Oxidation and reduction both occur at the cathode

C. Oxidation occurs at the cathode and reduction occurs at the anode

D. Oxidation occurs at the anode and reduction occurs at the cathode

Answer D

17. Write anode, cathode and cell reaction for the electrolysis of molten potassium chloride.

Explanation

$$KCl \xrightarrow{melting} K^+ + Cl^-$$

Anode reaction: $2Cl^- \rightarrow Cl_2 + 2e^-$

Cathode reaction: $2K^+ + 2e^- \longrightarrow 2K$

Cell reaction: $2K^+ + 2Cl^- \longrightarrow 2K + Cl_2$

18. Two steel needles dipped into dilute hydrochloric acid solution are connected to the terminals of a 9 volt battery. What is the needle connected to the end of the battery marked with a "+"?

Explanation: The electrode connected to the "+" terminal of a battery is anode.

19. Calssfiy the following as active or inert electrode.

a. Cu
b. Ag
c. Mg
d. Au
e. Pt
f. graphite
g. Zn

Answer

- b, d, e, and f are intert electrodes
- a, c and g are active electrodes.

20. During electrolysis, anions migrate toward anode and cations migrate toward anode. Why?

Explanation: This is because of unlike charge attract each other.

21. What is the difference between half-reactions and cell reaction?

Explanation:Half – reactions occur at electrodes, but cell reaction is the combination of half-reactions.

22. The electrodes that do not themselves involve in the redox reactions are called_____.

Answer: Inert electrodes

23. Write anode, cathode and cell reaction for the electrolysis o molten calciumchloride

Answer

$CaCl_2 \xrightarrow{\text{melting}} Ca^{2+} + 2Cl^-$

Anode reaction: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode reaction: $Ca^{2+} + 2e^- \longrightarrow Ca$

Cell reaction: $2Cl^- + Ca^{2+} \longrightarrow Ca + Cl_2$

24. Why electrolysis is a redox process?

*Explanation:*Because it is the combination of oxidation half-reaction and reduction half – reaction.

Applications of Electrolysis

- Industrially, electrolysis has the following applications.

Production of metals

- Group IA and IIA metals are active metals. There is no a better reducing agent to displace these metals. As a result, these metals are manufactured industrially by electrolysis.
- Aluminum is also manufactured industrially by **Hall process** using electrolysis.

ii Production of non-metal

➥ Chlorine, fluorine, oxygen and hydrogen are industrially produced by electrolysis.

iii Production of compounds

➥ Sodium hydroxide and potassium hydroxide are commercially produced from electrolysis of concentrated sodium chloride solution and potassium chloride respectively.

iv Electrorefing/purification of metals/

➥ Electrofining is an industrial electrolytic process in which a sample of impure metal acts as the anode and a sample of pure metal acts as cathode copper, silver, nickel and tin are purified by electrolysis.

During electrorefing:

Anode = impure metal

Cathode = pure metal

Electrolyte = salt of the metal

Example: purification of copper.

Electrolysis of copper (II) sulphate using copper electrode.

Anode = impure copper

Cathode = pure copper

Electrolyte = $CuSO_4$

Anode reaction: $Cu \longrightarrow Cu^{2+} + 2e$

Cathode reaction: $Cu^{2+} + 2e^- \longrightarrow Cu$

➥ Mass of anode decreases

➥ Mass of cathode increases

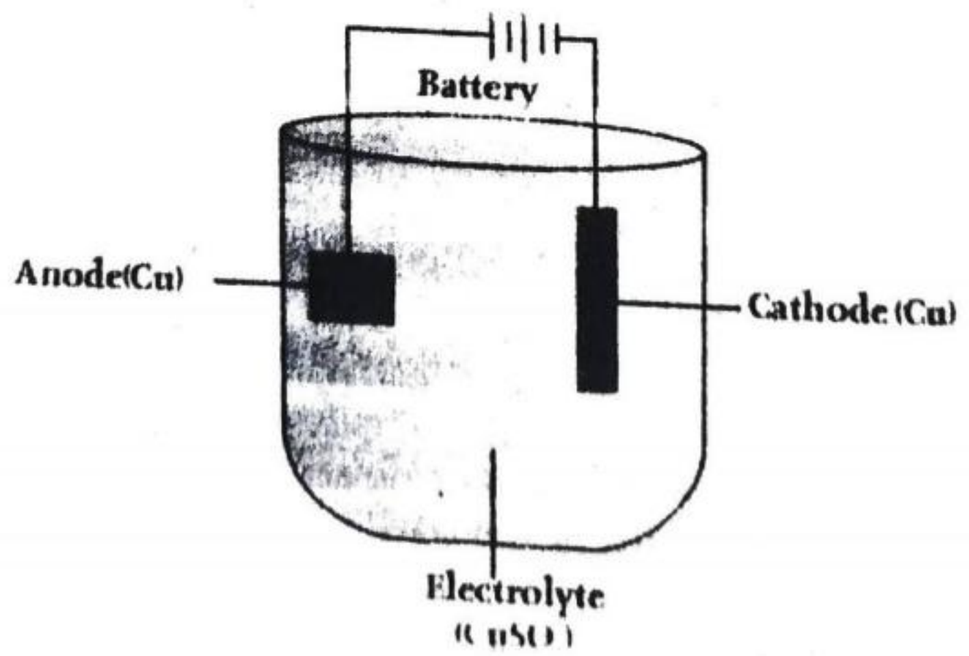


Figure 8.3 Purification of Copper

Electroplating

- Electroplating is the process that involves the coating of a solid surface with metal layer by the application of electrolysis

During electroplating:

Anode = plating metal

Cathode = material to be plated

Electrolyte = soluble salt of plating metal

Example

During electroplating of coin with gold,

Anode = gold

Cathode = coin

Electrolyte = solutions containing gold ions

Illustrative Examples

25. Suppose that you want to electroplate a spoon mode of iron with gold. Which of the following arrangements will make the process work out correctly? (EGSECE……..2003 E.C)

A. The electrolyte should be made of any soluble salts of iron.

B. The spoon should be made the cathode of the cell.

C. The electrolyte can be prepared from soluble salts of metals other than gold.

D. The cell should be constructed as a simple Galvanic cell.

Explanation

To electroplate a spoon made of iron with gold:

Anode = gold

Cathode = spoon

Electrolyte = soluble salt of gold

Electrochemical cell = electrolytic cell

Answer B

26. Which of the following is true about electrolytic refining of copper?
    A. The settling impurity may contain gold, silver and platinum
    B. The nature of electrolyte changes significantly
    C. The pure copper deposits at the anode
    D. The impure copper is used as cathode

Explanation:Anode = impure Cu, cathode = pure, Cu, pure Cu deposited at cathode. Electrolyte = $CuSO_4$.

Answer A

27. All of the following are the uses of electrolysis. Except
    A. Production of metals
    B. Production of non – metals
    C. Generate electricity
    D. Electroplating.

Explanation

Electrolysis consumes electricity but does not generate electricity.

Answer C

28. In an electroplating process
    A. The cell reaction is a spontaneous process
    B. The electrodes are made up of similar metallic substances
    C. The article to be plated is used as anode of the cell
    D. The electrolyte is salt of plating metal

Answer D

29. List some metals, non-metals and compounds that can be manufactured by electrolysis.

Answer

- Sodium, calcium, magnesium, aluminum – metals.
- Chlorine, hydrogen, oxygen, fluorine – non-metals
- Sodium hydroxide – compound

30. What is the relationship between amount of metal plated and amount of electricity delivered during electroplating ?

Explanation

➦ The amount of metal plated is directly related to the amount of electricity delivered.

31. During electrifying, why the mass of cathode increases?

***Explanation*:** The mass of cathode increases because pure metal is deposited on it?

32. The process of coating one metal with a thin layer of another metal, using electricity is called _____

Answer: Electroplating

8.3 Galvanic/Voltaic/Cells

Galvanic cells are electrochemical cells in which spontaneous redox reactions generate electricity.

➦ Chemical energy is converted into electrical energy.

In galvanic cells:

Anode = negative

Cathode = positive

There are three types of Galvanic cells.

These are:

1. Primary Galvanic cells
2. Secondary galvanic cell
3. Fuel cells

1. Primary galvanic cells

➦ They are non-rechargeable cells because it is impossible to regenerate the reactants.

➦ Their chemical reactions is irreversible.

Primary cells include:

1. The zinc – copper cell (Zn – Cu cell)

It is also called the **Daniel cell** because it was discovered by a British chemist **Daniel** in 1836.

Anode = Zinc in $ZnSO_4$ solution

Cathode = Copper in $CuSO_4$ solution

- It has anode and cathode compartement separated by in inverted u-tube filled with salt solution called **saltbridge.**

Anode reaction: $Zn \longrightarrow Zn^{2+} + 2e^-$

Cathode reaction: $Cu^{2+} + 2e^- \longrightarrow Cu$

Cell reaction: $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$

Voltage (V) = 1.10V

- As the zinc ions leave the solid zinc anode and enter the solution, electrons are lost behind the zinc anode, as a result, the zinc anode electrode aquire a negative charge.
- At the copper cathode, Cu^{2+} ions become attached to the electrode and seeks electrons to become reduced. This gives the copper electrode a positive charge.

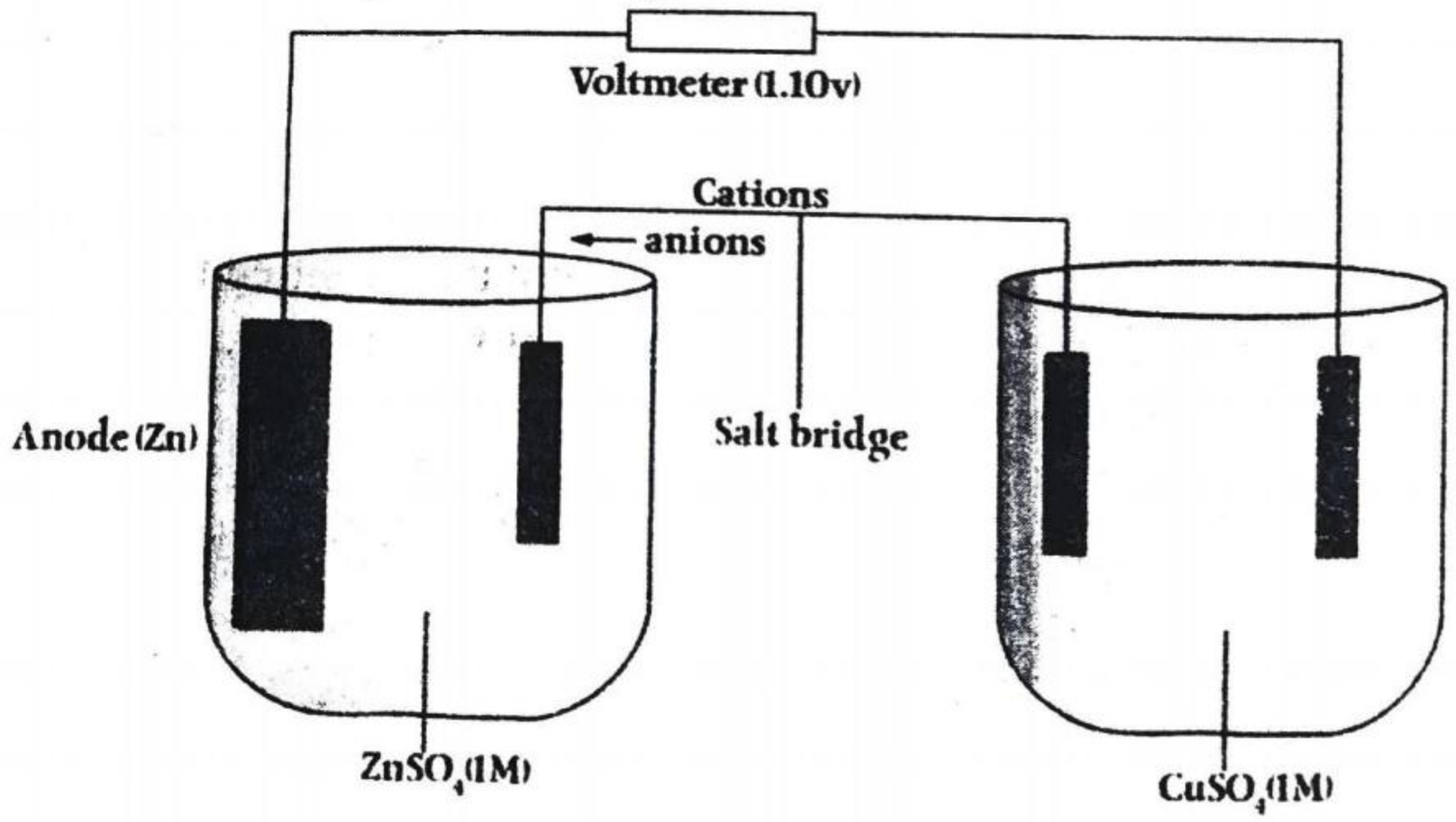


Figure 8.4 Daniel cell

- **A salt bridge serves three functions:**

1. It allow electrical contact between the two solutions
2. It prevents mixing of the electrode solutions.

3. It maintains the electrical neutrality in each half-cell as ions flow into and out of the salt bridge.

- A cell in which all reactants and products are in their standard states (1M or 1atm at 25°c) iscalled a **standard cell.**
- At standard conditions, the following experimental observations have been made about Daniel Cell.
  - The initial voltage is 1.10V
  - The mass of copper electrode increases and the concentration of Cu^{2+} in the solution around the copper electrode decreases as the cell operates.
  - The mass of zinc electrode deceases and concentration of Zn^{2+} around zinc electrode increases, as the cell operates.

2. The Leclanche dry cell (Zinc-carbon cell)

Anode = Zinc

Cathode = graphite

Electrolytes = NH_4Cl, $ZnCl_2$, MnO_2

Anode reaction: $Zn \longrightarrow Zn^{2+} + 2e^-$

Cathode reaction: $2NH^+_4 + 2MnO_2 \overset{+2e}{\rightarrow} Mn_2O_3 + 2NH_3 + H_2O$

Cell reaction: $Zn + 2NH_4^+ + 2MnO_2 \rightarrow Zn^{2+} + Mn_2O_3 + 2NH_3 + H_2O$

Voltage = 1.50V

- A build up of ammonia gas cannot be permitted to occur around the cathode since it would **disrupt the current.**

3. Alkaline battery

- More durable and expensive than Leclanche cell
- Anode = Zinc
- Cathode = MnO_2
- Electrolye = KOH or NaOH

Anode reaction: $Zn + 2OH^- \longrightarrow ZnO + H_2O + 2e^-$

Cathode reaction: $2MnO_2 + H_2O + 2e^- \longrightarrow Mn_2O_3 + 2OH^-$

Cell reaction: $Zn + 2MnO_2 \longrightarrow ZnO + Mn_2O_3$

Voltage = 1.5V

4. Silver oxide cells

➦ It is the most expensive battery

Anode = Zinc

Cathode = silver oxide

Electrolyte = NaOH or KOH

Anode reaction: $Zn + 2OH^- \longrightarrow Zn(OH)_2 + 2e$

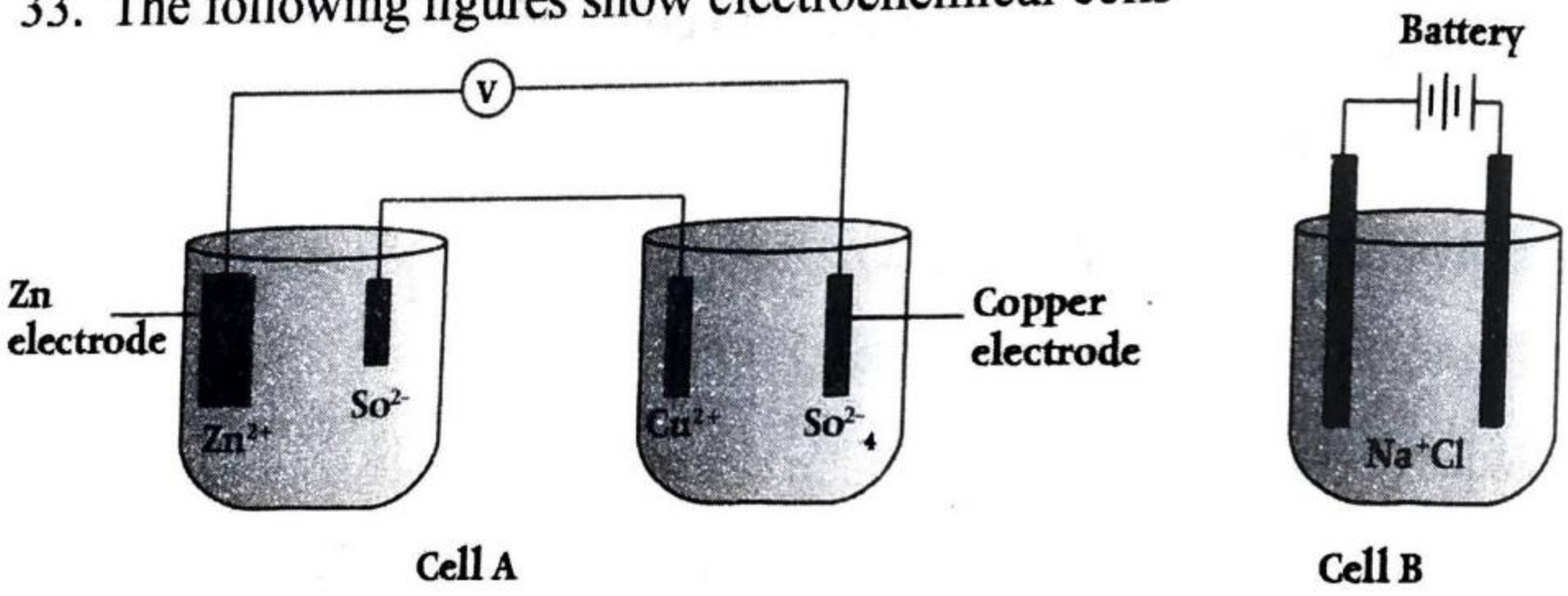
Cathode reaction: $Ag_2O + H_2O + 2e \longrightarrow 2Ag + 2OH^-$

Cell reaction: $Zn + Ag_2O + H_2O \longrightarrow Zn(OH)_2 + 2Ag$

Voltage = 1.5V

Illustrative examples

33. The following figures show electrochemical cells



Which of the following statements is true about these cells? (EGSECE… …….2007E.C)

A. Cell A converts chemical energy into electrical energy while cell B converts electrical energy into chemical energy.

B. In cell B elements flow from the negative terminal to the positive terminal of the battery in the external circuit.

C. In cell A electrons flow from Cu – electrode to Zn-electrode

D. Cell A converts electrical energy in to chemical energy while cell B converts chemical energy into electrical energy

Explanation

➦ Cell A is Galvanic cell and cell B is electrolytic cell. In both cells electrons flow from anode to cathode in the external circuit.

Answer A

34. A student collects the materials and equipments given below to construct a voltaic cell

i One strip of magnesium

ii 125mL of 0.20M $Mg(NO_3)_2$

iii 125ML of 0.20M $Cu(NO_3)$

iv Two 250mL beakers

v One strip of copper

vi Wire and a switch

35. Which additional item is required for the construction of the voltaic cell?

A. A cathode
C. An anode
B. A salt bridge
D. A battery

Explanation: A salt bridge is required to complete the circuit.

Answer: B

36. Which of the following is not an electrolyte in Leclanche cell?

A. H_2SO_4
C. MnO_2
B. NH_4Cl
D. $ZnCl_2$

Answer A

37. Which of the following generates small amount of voltage?

A. Leclanche cell
C. Silver oxide cell
B. Alkaline battery
D. Daniel cell

Explanation: Daniel cell generates 1.10V but the others generate 1.5V

Answer D

38. The anode used in most primary Galvanic cells is _____

Answer: Zinc

39. Classify the following as primary Galvanic cells, dry cell or wet cells.

a. Daniel cell
c. Alkaline battery
b. Leclanche cell
d. silver oxide cell

Explanation: Galvanic cells with aqueous electrolyte are wet cells while those with paste electrolytes are dry cells. Only Daniel cell is wet well.

40. How a salt bridge maintains electrical neutrality between the two electrodes solutions?

Explanation: The slat bridge releases negative ions into anode compartment and positive ions to the cathode compartment to maintain electrical neutrality between the two solutions.

41. What problem initiated scientists to discover dry cells?

Explanation: Wet cells such as Zn-Cu cell can serve as a source of electricity. However, they are not easily portable from one area to another area since they contain aqueous electrolyte.

42. Identify the negative and positive electrodes in Leclanche cell.

Answer

Negative electrode = Zinc

Positive electrode = graphite

2. Secondary Galvanic cells/reversible cells/

➥ They are rechargeable Galvanic cells

➥ Reactants are regenerated by the reverse reaction by passing a direct current through the cell in the direction opposite to the discharge current flow. This process is called **charging or recharging a cell or battery.**

Example of secondary Galvanic cells are:

i Lead storage battery/car battery/

➥ It is used in most automobiles

Anode = Pb

Cathode = PbO_2

Electrolyte = H_2SO_4

Anode reaction: $Pb(s) + SO_4^{2-}(aq) \rightleftharpoons^{discharge} PbSO_4(s) + 2e$

Cathodereaction: $PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) \rightleftharpoons^{(discharge)} pbso_4(s) + 2H_2O(1)$

Cell reaction: $Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq) \rightleftharpoons^{discharge} 2PbSO_4(s) + 2H_2O(1)$

V = 2.07v

The typical 12-V lead storage battery, therefore, contains **six 2-v cells**

connected in series.

ii **Nickel – cadmium battery (Nicad battery)**

It is used in battery – operated tools and portable computer.

Anode Cadmium (Cd)

Cathode Nickel (Iv) oxide (NiO_2)

Electrolyte = NaOH or KOH

- The voltage of Nicad battery is 1.40v.
- It has longer life than a lead stronge battery.
- It can be packed and sealed easily.

3. Fuel cells

- They are Galvanic cells that require continuous supply of reactants to keep functioning.
- Fuel cells donot store chemical energy like batteries.
- The common fuel cell is the **hydrogen oxygen** fuel cell
- In hydrogen – oxygen fuel cells.

- Anode and cathode = graphite
- Electrolyte = aqueous solution of base.

Anode reaction: $2H_{2(g)} + 4OH^-(aq) \longrightarrow 4H_{20}(1) + 4e^-$

Cathode reaction: $O_2(g) + 2H_2O(1) + 4e^- \longrightarrow 4OH^-(aq)$

Cell reaction: $2H_2(g) + O_2(g) \longrightarrow 2H_2O(1)$

Voltage = 1.23V

The cell is maintained at high temperature and the water produced by the cell reaction evaporates as it is formed.

Illustrative Examples

41. Which of the following is an example of a secondary voltaic cell?

    A. Zinc-carbon dry cell

    B. Lead-storage battery

    C. Daniel cell

    D. Leclanche cell

    Answer B

42. Which of the following electrochemical cell is used in automobiles? (EGSECE….2006E.C)

    A. Ni-cad battery

    B. Fuel cell

    C. Daniel cell

    D. Lead – storage battery

    Answer D

43. Which of the following processes uses a Galvanic cell?

    A. Refining impure copper

    B. Lead storage battery

    C. Electrolysis of fused sodium chloride

    D. Hall procell production of aluminium

    Answer B

44. Consider the following electrochemical reaction in lead storage battery

    $Pb + PbO_2 + 2H_2SO_4 \longrightarrow 2PbSO_4 + 2H_2O$

    Which species is reduced when the battery is discharging? (EGSECE… .2001E.C)

    A. $PbSO_4$

    B. PbO_2

    C. H_2SO_4

    D. Pb

    Explanation: The oxidation state of Pb in PbO_2 is +4 while in $PbSO_4$ is +2. Thus, the reduced substance is PbO_2.

    Answer B

45. Write the anode, cathode and cell reaction during charging of a battery.

Explanation: Anode reaction: $PbSO_4 + 2H_2O \rightarrow PbO_2 + 4H^+ + SO^{2-}_4 + 2e^-$

Cathode reaction: $PbSO_4 + 2e^- \longrightarrow Pb + SO_4^{2-}$

Cell reaction: $2PbSO_4 + 2H_2O \longrightarrow PbO_2 + Pb + 2H_2SO_4$

46. What is the difference between fuel cells and batteries?

Explanation: Batteries store chemical energy but fuels donot store chemical energy.

Solved problems

47. Write the difference between electrolytic and Galvanic cells

Explanation

Electrolytic cell	Galvanic cell
➦ Convert electrical energy into chemical energy ➦ Non-spontaneous rodox reaction ➦ Anode is positive and cathode is negative	➦ Convert chemical energy into electrical energy ➦ Spontaneous redox reaction ➦ Anode is negative and cathode is positive

48. During electrolysis of fused KF:

a. Identify the charge carriers
b. Write anode half-reaction
c. Write cathode half-reaction
d. Write the cell reaction
e. Identify the products formed at cathode and anode

Answer

a. K^+ and F^-
b. $2F^- \longrightarrow F_2 + 2e$
c. $K^+ + e^- \longrightarrow K$
d. $2K^+ + 2F^- \longrightarrow 2K + F_2$
e. F_2 is librated at anode and K is deposited at cathode.

49. The commercial production of magnesium involves the electrolysis of molten $MgCl_2$ why cannot the simpler electrolysis of $MgCl_2$(aq) be used instead?

Explanation: In molten $MgCl_2$ no selection for cathode but in aqueous $MgCl_2$ there is a chance for the cathode to select H^+ instead of Mg^{2+}.

50. Write the difference between metallic and electrolytic conduction.

Answer

Metallic conduction	Electrolytic conduction
➦ No chemical change	➦ There is chemical change
➦ Transmit in all states	➦ Transmit in molten and aqueous state

51. What are the charge carriers in metallic conductors?

Answer

Free or valence electrons of metals

52. What are the products formed during electrolysis of molten magnesium chloride?

Answer: Mg and Cl_2

53. Do you think that primary voltaic cell is rechargeable?

Answer : No

54. A voltaic cell is constructed with Pb(s) in $Pb(NO_3)_2$(aq) for other cell. If Zn is a better reducing agent than Pb and the salt bridge contains an aqueous solution of KNO_3, answer the following questions.

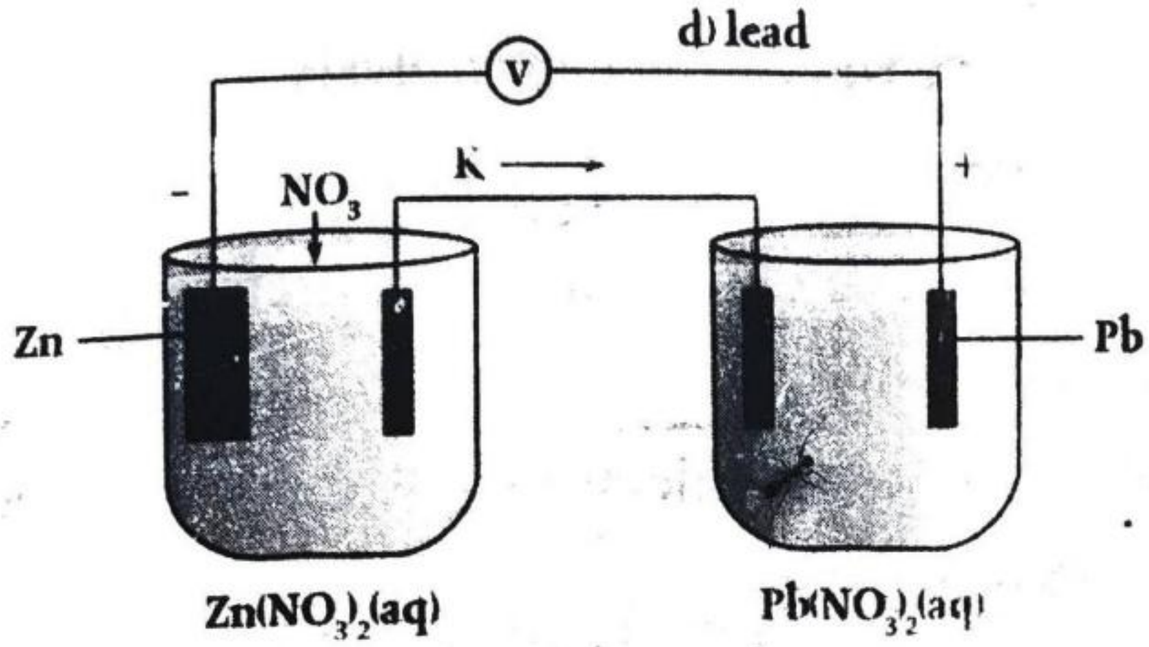
a. Identify the anode and cathode
b. Which ions move towards the anode
c. Which ions move towards the cathode
d. Which metal is positive electrode
e. Draw the voltaic cell

Explanation

a. More reactive metal is anode and less reactive metal cathode. Zn is anode and Pb is cathode.

b. NO_3^- from the salt bridge moves into anode compartment to neutralize the excess Zn^{2+} formed

c. K^+ moves into cathode compartment to replace the charge of lead ion (Pb^{2+}) reduced.



Supplementary Exercise

I. Choose the best answer from the given alternative

1. Which of the following set of application is associated with electrolysis?
   A. Extraction of metals, electroplating, batteries.
   B. Extraction of metals, electroplating, purification of metals.
   C. Fuel cells, purification of metals, electroplating
   D. Electroplating, batteries, fuel cells
2. Which of the following species is preferentially discharged at the anode in the electrolysis of brine?
   A. OH^-
   B. H_2O
   C. Cl^-
   D. Na^+
3. In an electrolysis experiment, a student observed that the mass of anode decreases with the corresponding increase in the mass of cathode. Which of the following were the appropriate electrolytic solution and electrodes for the experiment?
   A. Sodium hydroxide and carbon electrodes
   B. Copper (II) sulphate and copper electrodes
   C. Copper (II) sulphate and platinum electrodes
   D. Lead bromide and copper electrode.
4. Which of the following is true about electrolysis?

A. It is a process in which chemical energy is used to produce electrical energy.

B. Electrodes are used to block the transfer of electrons during electrolysis

C. Cathode attracts cations during electrolysis

D. The concentration of the ions does not have effect on electrolysis

5. The selection of an ion by an electrode in preference to other ions during electrolysis depends on

A. The magnitude of the charge on the ion

B. The molar mass of the ion

C. The external electromotive force applied to the cell

D. The nature of electrodes used

6. If molten $CuSO_4$ is electrolyzed using platinum cathode and anode, which half-reaction takes place at the cathode?

A. $Cu^{2+} + 2e \longrightarrow Cu$

B. $S^{2-} \longrightarrow S + 2e^-$

C. $Cu \longrightarrow Cu^{2+} + 2e^-$

D. $S + 2e^- \longrightarrow S^{2-}$

/. Which of the following is wet cell?

A. Leclanche cell

B. Silver oxide cell

C. Lead storage battery

D. wrist watch battery

8. Which one of the following statements is correct about the electrolysis process of aqueous solutions of electrolytes?

A. Reduction takes place at the cathode

B. Electrons flow from negative to positive electrode in the external circuit

C. The negative electrode discharges one anion

D. Positive ions migrate towards anode

9. By which method can be metal obtained from its molten salt?

A. Analysis

B. Reduction

C. Oxidation

D. distillation

10. Which of the following groups of metals is extracted by electrolysis method?

A. Au, Ag, Cu
C. Al, Mg, Na
B. Mg, Cu, Fe
D. K, Ca, Fe

11. Which compound in its liquid state transmits electricity?

A. Silcon dioxide
C. carbon monoxide
B. Sugar
D. calcium iodide

12. The electrolyte used in car battery is

A. H_2SO_4
C. HCl
B. HNO_3
D. CH_3COOH

13. All of the following are strong electrolytes except.

A. $HNO_3(aq)$
C. NaOH(aq)
B. NaCl(aq)
D. $H_2CO_3(aq)$

14. Which of the following would discharge one mole when 96500 coulombs of electricity passes through their solution?

A. Ca^{2+}
C. Ag^{+}
B. Al^{3+}
D. S^{2-}

15. Which of the following is not primary galvanic cell?

A. Daniel cell
C. Alkaline battery
B. Cooperative cell
D. Nicad battery

16. The difference between a galvanic and electrolytic cell is that unlike electrolytic cell, the galvanic cell

A. Is used in electroplating process
B. Is used in production of metals from their ores
C. Has spontaneous redox reaction
D. Uses electrical energy to bring about a chemical change.

17. An electrolytic cell contains copper chloride solution. When direct current passes through the solution the following reaction takes place:

$CuCl_2(aq) \longrightarrow Cu + Cl_2$

Which of the following is correct about this reaction?

A. Electrons enter the solution at the anode
B. Electrons are taken up by the negative ions
C. Chlorine gas is librated at the anode
D. Reduction occurs at the anode

18. In galvanic cell, oxidation occurs at
A. Cathode
B. Anode
C. salt bride
D. Electrolyte

19. When an electric current is passed through a certain substance, magnesium and chlorine are formed. This substance cannot be
A. An element
B. A compound
C. a homologeneous mixture
D. none

20. Why must a salt bridge or semi permeable membrane be used to separate the two half-cell reactions in an electrochemical cell?
A. It prevents the mixing of ions in the two half-cells.
B. It allows electrical neutrality to be maintained by migration of ions from one half cell to the other
C. It allows the mixing of oxidized and reduced species.
D. It allows electrons to flow between the two half-cells.

21. When a lead storage battery is charged, lead sulfate is:
A. Formed at the cathode
B. Formed at the anode
C. Formed at both electrodes
D. Removed from both electrodes

22. Which one of the following solution shows no current flow in an electronic cell?
A. Molten solution of sugar
B. Molten sodium nitrate
C. Sulfuric acid solution
D. Sodium hydroxide solution

23. Metals conduct electricity. This is because metals posses
A. Freely moving ions
B. Inner electrons held in fixed position
C. Delocalized valence electrons
D. Valence electrons that are strongly bound to the molecules

24. The electrolyte in alkaline battery is

A. NH_4Cl
B. H_2SO_4
C. MnO_2
D. KOH

25. Increasing the concentration of ions in an electrolyte solution.

A. Increases the extent of conduction through it.
B. Decreases the extent of conduction through it.
C. Has no effect on the conduction of electricity
D. Changes the direction of electron flow

II. Short answer

26. Write the difference between

a. Cation and anion
b. Anode and cathode
c. Inert and active electrode
d. Strong electrolyte and weak electrolyte

27. Classify the following as electrolytes or non-electrolytes

a. Trichloromethane ($CHCl_3$)
b. Sugar ($C_{12}H_{22}O_{11}$)
c. Ethanol (C_2H_5OH)
d. Carbonic acid (H_2CO_3)
e. Table salt (NaCl)
f. Ammonia (NH_3)

28. _____ is a solution or a molten compound which cannot be decomposed by an electric current.

29. _____ are two poles of graphite or metal at which electrons enter or leave an electrolyte.

30. Given the cell reaction:

$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$, write the anode and cathode half reactions.

UNIT 9

Chemistry industry and Environmental pollution

Unit Outline:

9.1 Natural Resources

9.2 Production of some Important Metals

9.3 Production of some Important non - metals

9.4 Some Industries in Ethiopia

9.5 Environmental Pollution

Introduction

A chemical industry is an institution that produces new chemical substance through chemical processes.

9.1 Natural resources

Natural resources are resources that exist in nature.

Natural resources are classified into two. These are:

1. Renewable natural resources
2. Non-renewable natural resources

1. Renewable natural resources

- They are replaced by natural process

Example

- animals
- Soil
- water
- Plants
- Fire wood

2. Non-renewable natural resources

- They cannot be replaced by natural processes.

Example:

- Petroleum
- Natural gas
- Minerals
- Coal

Illustrative examples

1. Of the following substances, which one in NOT a natural resource? (EGSECE……2008E.C)

   A. Metal ore
   C. Glass
   B. Soil
   D. stone

***Explanation*:** Glass is made by man in industry. Industry products are artificial resources.

Answer C

2. Which of the following natural resources is non-renewable?

   A. Petroleum
   C. Plants
   B. Water
   D. animals

Answer A

Uses of Iron

Iron is the cheapest metal and in the form of steel, the most useful. There are two types of steels, **carbon steel and alloy steel.**

- Carbon steel can be classified as **mild steel, medium steel or high carbon steel.**

Types of steel	Percentage of carbon	Uses
Mild steel	≤ 0.2	General engineering: motor car bodies, wire, piping nuts and bolts
Medium steel	0.3 – 0.6	Beams and girders, springs
High carbon steel	0.6 – 1.5	Drill bits, knives, hammers, chisels.

- **Alloy steel** is also classified as **stainless steel and tool steel**.
- Stainless steel = Fe + Cr + Ni
- Tool steel = Tungsten steel and manganese steel.
- Alloy steel is used in rock drills, cutting edges, and parts of machinery.

Illustrative Examples

13. What is the role of calcium carbonate in the blast furnace during the extraction of iron? (EGSECE-2008E.C)

A. Used as a raw material

B. To facilitate the removal of impurities

C. To reduce iron ore to iron

D. To oxidize carbon to carbonmonoxide

Explanation:Calcium carbonate is used to remove impurities in the form of slag.

Answer B

14. Among the following, identify the purest form of iron

A. Wrought iron

B. Cast iron

C. steel

D. pig iron

Answer A

3. Which of the following natural resources is non-renewable?

A. Water
B. Soil
C. Natural gas
D. Plants

Answer C

4. Minerals are .....

A. Used as raw materials in chemical industries
B. Industrial products
C. Replenished with the same rate as they are consumed
D. Renewable resources

Explanation:Minerals are non-renewable natural resources used as raw materials in chemical indstries.

Answer A

5. Which of the following is not true about chemical industries? Chemical industry:

A. Converts reactants into finished and semi-finished products.
B. Uses naturally existing raw materials to produce desired products.
C. Uses either natural resources or other substances as raw material.
D. Focuses always on chemical reactions that release heat energy.

Answer D

6. List some general characteristics of chemical industries

Answer

- They use naturally – available raw materials
- They involve chemical reactions to transform raw materials into finished and semi-finished products.
- They consume relatively large quantities of energy and water
- Unless controlled, they release wastes to the environment in the form of gas(smoke), liquid or solid.

9.2 Production of some important metals

Aluminum (Al)

Occurrence

- It does not exist free in nature
- Its main ore is **bauxite** ($Al_2O_3.2H_2O$)
- Other minerals containing aluminum are:
  - Orthoclase – $KAlSi_3O_6$
  - Beryl – $Be_3Al_2Si_6O_{18}$
  - Cryolite – Na_3AlF_6
  - Corundum – Al_2O_3

Extraction

- Aluminum is industrial manufactured with **Hall process** by using electrolysis from bauxite.
- **Cryolite** is added to bauxite to lower its melting point.

Cathode reaction: $4Al^{3+} + 12e^- \longrightarrow 4Al(1)$

Anode reaction: $6O^{2-} \longrightarrow 3O_2 + 12e^-$

$$3C(s) + 3O_2(g) \longrightarrow 3CO_2(g)$$

Overall reaction: $4Al^{3+} + 6O^{2-} + 3C(s) \longrightarrow 4Al(1) + 3CO_2(g)$

Physical properties of aluminum

- It is a soft, white metal
- It is a good conductor of heat and electricity.
- It has a brilliant luster.
- It is malleable and ductile.

Chemical properties of aluminum

- Aluminum reacts with non-metal

$$4Al(s) + 3O_2(g) \longrightarrow 2Al_2O_3(s)$$

$$4Al(s) + 6S(s) \longrightarrow 2Al_2S_3(s)$$

$$2Al(s) + N_2(g) \longrightarrow 2AlN(s)$$

$$2Al(s) + 3F_2(g) \longrightarrow 2AlF_3(s)$$

➦ Aluminum is amphoteric metal, reacts with acids and bases

$$2Al(s) + 6HCl(aq) \longrightarrow 2AlCl_3(aq) + 3H_2(g)$$

$$2Al(s) + 2NaOH(aq)H_2O + 6H_2O(1) \rightarrow 2NaAl(OH)_4(aq) + 3H_2(g)$$

Aluminum is passive towards nitric acid due to a thin transparent film of Al_2O_3 formed when aluminum comes into contact with air.

Uses of Aluminum

- ➦ Its alloys, duralumin (Al/Mg/Cu) and Magnalium (Al/Mg) are light, strong and corrosion resistant.
- ➦ It is used to make cooking utensils, brewing vats, food procelling vessels and packing and copping milk bottles.
- ➦ It is used for making electrical conductors for transmission lines.
- ➦ The very high amount of heat released by the thermite reaction makes it useful in welding large steel parts such as propeller shafts and rudderposts on ships and heavy machinery.

  $2Al + Fe_2O_4 \rightarrow Al_2O_3 + 2Fe$ thermite reaction
- ➦ It is used as reducing agent for small scale extraction of chromium

Illustrative Examples

7. Hall's process of production of aluminum involves (EGSECE... ..2004E.C)

   A. Heating cryolite with CO
   B. Electrolytic reduction of Al_2O_3
   C. Displace Al from its salts using active metals
   D. Reducing $AlCl_3$ using coke

Explanation:Aluminum is not manufactured by chemical reduction using other reducing agents.

Answer B

8. Which property of aluminum makes it preferably best for construction of air planes frame work?

A. High ductility
B. Good electrical conductivity
C. Resistance to corrosion
D. Lightness and strength

Explanation

Uses of aluminum	Reason
➦ Construction of airplanes	➦ Lightness and strength
➦ Making cooking utensils	➦ Good conductivity of heat
➦ Electric cables	➦ Light, ductile, corrosion resistance

Answer D

9. Which of the following elements is the third most abundant element in the earth's crust?

A. Silicon
B. aluminum
C. Calcium
D. iron

Explanation

Abundance of elements in the earth's crust is oxygen > silicon > aluminum > iron > . . . .

Answer B

10. What is the purpose of crylote in the production of aluminum by Hall process?

Explanation

It is used to lower the melting point of bauxite.

11. State the most important alloys of aluminum

Answer:

➦ Dulalumin = Al + Mg + Cu
➦ Magnalium = Al + Mg
➦ Aluminum bronze = Al + Cu

12. What is thermite reaction?

Explanation

Thermite reaction is the reaction between aluminum and iron (III) oxide

$2Al + Fe_2O_3 \longrightarrow 2Fe + Al_2O_3$, $\Delta H = -84KJ/mol$

ii Iron (Fe)

Occurrence

- It does not exist free in nature
- Its main ore is hematite (Fe_2O_3)

Other minerals of iron are:

- **Limonite – $Fe_2O_3 . H_2O$**
- **Magnetite – Fe_3O_4**
- **Siderite – $FeCO_3$**
- **Iron pyrite or fool's gold – FeS_2**

Extraction

- Iron is industrially manufactured in blast furnace

Raw materials: iron ore – source of iron, lime stone – remove impurities, hot air – source of heat, coke – reducing agent.

Chemical reduction of hematite is shown below

$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$

$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$

$FeO + CO \longrightarrow Fe + CO_2$

- The iron obtained directly from the blast furnace is known as **Pig iron.**
- The purest form of iron is called **Wrought iron.**

Conversation of Pig Iron to Steel

- Pig iron is converted into steel by three processes

The Bessemer Converter

- This process requires only 15minutes.
- The quality of steel produced by this process is variable
- The reaction is difficult to control
- Uses $CaCO_3$ and $MgCO_3$

The open – hearth furnace

- Lined with MgO and CaO.
- The process requires 8-10 hours
- More easily controlled reaction
- High quality steel is formed.

Basic oxygen process

- Replaced the open hearth furnace because of its high speed.
- It is expensive.
- It requires only about 20 to 25 minutes.

Physical properties of Iron

- It is grey, luster, malleable and ductile metal.
- It is good conductor of heat and electricity.
- It is heavy metal with density $7.87g/cm^3$
- It is ferromagnetic metal meaning permanently magnetized.

Chemical properties of iron

- Iron reacts with air

$4Fe + 3O_2\ Fe_2O_3 \cdot xH_2O$

- It reacts with acids

$Fe + 2HCl \longrightarrow FeCl_2 + H_2$

$2Fe + 6H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + 3SO_2 + 6H_2O$

- It reacts with non-metals such as sulphur and chlorine

$Fe + S \longrightarrow FeS$

$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

Uses of Iron

Iron is the cheapest metal and in the form of steel, the most useful. There are two types of steels, **carbon steel and alloy steel.**

- ➦ Carbon steel can be classified as **mild steel, medium steel or high carbon steel.**

Types of steel	Percentage of carbon	Uses
Mild steel	≤ 0.2	General engineering: motor car bodies, wire, piping nuts and bolts
Medium steel	0.3 – 0.6	Beams and girders, springs
High carbon steel	0.6 – 1.5	Drill bits, knives, hammers, chisels.

- ➦ **Alloy steel** is also classified as **stainless steel and tool steel** .
- ➦ Stainless steel = Fe + Cr + Ni
- ➦ Tool steel = Tungsten steel and manganese steel.
- ➦ Alloy steel is used in rock drills, cutting edges, and parts of machinery.

Illustrative Examples

13. What is the role of calcium carbonate in the blast furnace during the extraction of iron? (EGSECE-2008E.C)

A. Used as a raw material

B. To facilitate the removal of impurities

C. To reduce iron ore to iron

D. To oxidize carbon to carbonmonoxide

Explanation:Calcium carbonate is used to remove impurities in the form of slag.

Answer B

14. Among the following, identify the purest form of iron

A. Wrought iron

B. Cast iron

C. steel

D. pig iron

Answer A

15. What is the main ore of iron?

Answer: hematite (Fe_2O_3)

16. Complete and balance the following

a. $Fe + O_2 \longrightarrow$

b. $Fe + HCl \longrightarrow$

c. $Fe + Cl_2 \longrightarrow$

d. $Fe + S \longrightarrow$

e. $Fe + H_2SO_4 \longrightarrow$

Answer

a. $4Fe + 3O_2 \longrightarrow 2Fe_2O_3 . x\ H_2O$

b. $Fe + 2HCl \longrightarrow FeCl_2 + H_2$

c. $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

d. $Fe + S \longrightarrow FeS$

e. $Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2$

17. What are the two common oxidation numbers of iron?

Solution:

➦ Iron (II) and Iron (III)

Iron (II) compounds have pale – green color and iron (III) compounds have yellowish brown color.

18. What are the component metals for stainless steel?

Answer: Iron, chromium and nickel

19. Pig iron is much cheaper than steel but is not used in construction. Explain why pig iron is not suitable for erecting buildings and bridges.

Explanation:The carbon content of pig iron is too high, it makes pig iron relatively hard and brittle.

20. What is the purpose of coke in the blast furnace?

Solution:

It is reducing agent.

21. Write the balanced acid – base reaction that occurs in the blast furnace.

Solution:

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

22. Write the components of Pig iron.

Solution:

Pig iron contains about 4% carbon, 2% silicon, a trace of sulphur and 1% of phopherus and manganese. Sulphur is probably the worst impurity (making steel break when worked) and must be avoided.

iii Copper (Cu)

Occurrence

- Copper exists both in free and combined states
- Its main ore is **chalcopyrite ($CuFeS_2$)**
- Other minerals of copper are:
  - Chalcocite ……Cu_2S
  - Covellite……CuS
  - Cuprite …… Cu_2O
  - Malachite……($CuCO_3Cu(OH)_2$)
  - Tenorite……..CuO

Extraction

A feal of the oxide ores of copper can be reduced directly to the metal by heating with coke, but the bulk production is from sulphide ores, particularly chalcopyrite. The crushed and ground sulphide ore is first concentrated by flotation and then roasted in a limited supply of air

$$2CuFeS_2(s) + 4O_2(g) \longrightarrow Cu_2S + 2FeO + 3SO_2$$

- Limestone and sand are added to the mixture to form a molten slag that removes many impurities. For example, iron (11) oxide

$$FeO + SiO_2 \longrightarrow FeSiO_3$$

The Cu_2S is then smelted or reduced to copper by heating in a limited amount of air.

$$CuS(s) + O_2(g) \longrightarrow 2Cu(s) + SO_2(g)$$

- The copper produced by the above process is called **blister copper** and is 96-99.5% copper and contains iron and silver impurities.
- Copper is purified by using **electrolysis**.

The steps in the production of copper in short is: concentrating ore by floation→ smelting → reduction → electro refining.

Physical properties of copper

- It is malleable and ductile.
- It is very good conductor of heat and electricity.
- It is heavy metal with a density of 8.96g/cm3
- It has high melting and boiling point.

Chemical properties of copper

- Copper cannot displace hydrogen from dilute acids, however, it can be oxidized by oxidizing acids such as dilute and concentrated sulphuric acid.

$$3Cu(s) + 8HNO_3\,(aq) \longrightarrow 3Cu(NO_3)_2\,(aq) + 2NO(g) + 4H_2O(l)$$

$$Cu(s) + 2H_2SO_4(conc.) \longrightarrow CuSO_4\,(aq) + SO_2(g) + 2H_2O(1)$$

- Copper corrodes in moist air as a result of oxidation caused by a mixture of water, oxygen and carbondioxide

$2Cu(s) + H_2O(1) + O_2(g) + CO_2(g) \longrightarrow Cu_2(OH)_2CO_3(s)$ basic copper Carbonate

Use of copper

- Copper is widely used in the electrical industry.
- It is also used to make alloys such as brass (Cu/Zn), bronze (Cu/Sn) and copper nickel (Cu/Ni)
- Brass is used for hard ware taps, terminals and pipes
- Bronze is used to make coins, medals, bells and machinery parts.

Illustrative Examples

23. Which of the following is the correct sequence for the industrial production of pure copper from chalcophyrite ($CuFeS_2$)? (EGSEC E.....2007E.C)

A. Roasting of the are⟶ smelting of the ore in lime stone ⟶ concentrating the ore ⟶ reducing the ore in limited oxygen ⟶ electrolytic refining.

B. Concentrating the ore ⟶ smelting of the ore in lime stone ⟶ roasting of the ore in oxygen ⟶ reducing the eore in limited oxygen ⟶ electrolytic refining

C. Smelting of the ore⟶ roasting of the ore in oxygen⟶reducing the ore in limited oxygen ⟶ electrolytic refining

D. Concentrating the ore ⟶ Roasting of the ore in oxygen ⟶ smelting of the ore in limited oxygen ⟶ electrolytic refining

Answer D

24. Which one of the following elements is the component of both brass and bronze?

A. Ni B. Zn C. Sn D. Cu

Answer D

25. Which of the following is the primcipal ore of copper?

A. $CuFeS_2$ C. CuO

B. Cu_2S D. $CuCO_3$

Answer A

26. Write the disproportion reaction of the unstable Cu^{+1}.

Answer: $2Cu^{+}(aq) \longrightarrow Cu^{2+}(aq) + Cu(s)$

27. What is the color of copper (1) and copper (II) compounds?

Solution

➦ Copper (I) compounds are colorless while copper (II) compounds are blue in color.

28. What is the metal that turns green due to the formation of verdigris?

Solution: Copper

9.3 Production of some important non-metals

i Nitrogen

Occurrence

- Nitrogen occurs free in the atmosphere as simple gaseous diatomic molecules, N_2. About 78% of air by volume is nitrogen.
- It also occurs in plants and animals in the form of proteins.
- The most important mineral sources of nitrogen are **salt peter (KNO_3) and chile salt peter ($NaNO_3$).**

Physical properties of nitrogen

- It is a colorless, odourless and tasteless gas.
- It is slightly less dense than air
- It boils at -196^0c

Chemical properties of nitrogen

- Nitrogen is less reactive non-metal due to the nitrogen – nitrogen triple bond.
- It reacts with very active metals.

$$6Li(s) + N_2(g) \longrightarrow 2Li_3N(s)$$

- It reacts with hydrogen and oxygen at high temperatures.

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

$$N_2(g) + O_2(g) \longrightarrow 2NO(g)$$

Preparation of Nitrogen

- Industrially, nitrogen is manufactured by **fractional distillation of liquid air.**
- In the laboratory, nitrogen is prepared by thermal decomposition of ammonium nitrite:

$$NH_4NO_2(s) \longrightarrow 2H_2O(1) + N_2(g)$$

Uses of nitrogen

- It is used to create inert atmosphere.
- Liquid nitrogen is used as a refrigerant for preserving blood, bull's semen and some foods.
- The most important use of nitrogen is in the production of ammonia, NH_3

Illustrative Examples

29. The most abundant gas in the atmosphere is

A. N_2 B. O_2 C. CO_2 D. Ar

Explanation: 78% by volume of the air is nitrogen.

21% by volume of the air is oxygen.

Answer: A

30. Which of the following does not describe the chemical properties of nitrogen? (EGSECE……2008E.C)

A. It is a relatively reactive gas
B. It combines with hydrogen to form ammonia
C. Reacts with group IA and IIA
D. It reacts with oxygen

Answer A

31. Which of the following is not the uses of nitrogen?

A. Creating inert atmosphere
B. For preserving blood and some foods
C. Micronutrient for plants
D. Making aminoacids

Answer C

32. It is difficult to imagine a living system that does not contain nitrogen. Why?

Explanation: The reason for this is nitrogen is an essential component of proteins, nucleic acids, vitamins and hormones that make life possible.

33. Nitrogen is less reactive element, why?

Explanation: It is due to the nitrogen – nitrogen triple bond (:N≡N:)

ii **Phosphorus (P)**

Occurrence

- ➦ It does not found free in nature.
- ➦ It occurs in combined state as:
  - Calcium phosphate – $Ca_3(PO_4)_2$
  - Fluorapatite – $Ca_5(PO_4)_3F$
- ➦ Its compounds are found in living things.

Physical properties of phosphorus

- ➦ Phosphorus has two allotropes. White phosphorus and red phosphorus.

Table 9.1. Different properties of white phosphorus and red phosphorus.

White phosphorus	Red phosphorus
➦ Boils at 287°c	➦ Sublimes at 416°c
➦ Density 1.8g/m²	➦ Density 2.16g/mL
➦ White waxy soldi	➦ Dark red powder
➦ Glows in dark	➦ Does not glow in the dark
➦ Soluble in carbondisulphere(CS_2)	➦ Not soluble in carbon disulphide (CS_2)
➦ Stored in water	➦ Stored in CS_2
➦ Extremely poisonous	➦ Relatively non-toxic compared to white phosphrous

Chemical properties of phosphorus

- ➦ It reacts with metals

$$6Ca(s) + P_4(s) \longrightarrow 2Ca_3P_2(s)$$

- ➦ It reacts with non-metals

$$P_4(s) + 3O_2(s) \longrightarrow P_4O_6(s)$$

$$P_4(s) + 5O_2(g) \longrightarrow P_4O_{10}(s)$$

$$4P_4(s) + 5S_8(g) \longrightarrow 4P_4S_{10}(s)$$

$$8P_4(s) + 3S_8(g) \longrightarrow 8P_4S_3(s)$$

$$P_4(s) + 6Cl_2(g) \longrightarrow 4PCl_3(s)$$

$$P_4(s) + 10Cl_2(g) \longrightarrow 4PCl_5(s)$$

Preparation of phosphorus

White phosphorus can be obtained by heating calcium phosphate with coke and silica /sand/ at very high temperature.

$$2Ca_3(PO_4)_{2(s)} + 10C(s) + 6SiO_2(s) \longrightarrow 6CaSiO_3(s) + 10CO(g) + P_4(g)$$

Red phosphorus is prepared by heating white phosphorus in the absence of air at about 300^0c for several days.

$$nP_4 \longrightarrow (P_4)_n$$

white phosphorus Red phosphc

Uses of phosphorus

- Phosphorus is used to make matches.
- White phosphorus is used to make phosphoric acid.
- It is used to make fireworks, smoke bombs, rat poisons and tracer bullets.
- It is essential for plant growth and to make fertilizers.

Illustrative Examples

34. Which of the following elements, occurs in nature only in combined form? (EGSECE……2000E.C)

A. Nitrogen
B. Phosphorus
C. Sulphur
D. Carbon

Answer B

35. Which of the following is not the purpose of phosphorus?

A. It is used to make fertilizer
B. It is used to make matches.
C. It is used to make sulphuric acid.
D. It is used to make rat poison.

Answer C

36. What are the two important allotropes of phosphorus? Which one is more stable.

Solution

- White and red phosphorus. Red phosphorus is more stable than white phosphorus.

37. Define allotropes

Solution

- Allotropes are different forms of the same element with the same physical state and isotopic composition but different physical properties, structure and reactivity.

38. Complete and balance the following equation.

a. $P_4 + Li \longrightarrow$ b. $Ca + P_4 \longrightarrow$

Answer:

a. $P_4 + 12Li \rightarrow 4Li_3P$ b. $6Ca + P_4 \rightarrow 2Ca_3P_2$

iii **Oxygen**

Occurrence

- **Oxygen** is the most abundant element on earth.
- The earth's crust is 46.6% oxygen by mass, the atmosphere contains 21% molecular oxygen by volume and the oceans contain 86% oxygen by mass.

Pereparation of Oxygen

- Industrially, oxygen is manufactured by **fractional distillation of liquid air.**
- In the laboratory oxygen can be prepared by the following methods.

1. By decomposition of hydrogen peroxide

$$2H_2O_2\,(aq) \longrightarrow O_2(g) + 2H_2O(1)$$

2. By decomposition of potassium chlorate.

$$2KClO_3(s) \longrightarrow 2KCl(s) + 3O_2(g)$$

3. By the decomposition of mercury (II) oxide.

$$2HgO(s) \rightarrow 2Hg(1) + O_2\,(g)$$

Physical properties of oxygen

- Oxygen is colorless, odorless and tasteless gas at room temperature.
- It has two allotropes, these are oxygen (O_2) and ozone (O_3)

Chemical properties of Oxygen

Oxygen reacts with almost all elements except noble gases and inert metals such as gold, platinum, palladium etc.

Example:

$$2Ca + O_2 \longrightarrow 2CaO$$

$$2C + O_2\,(\text{limited}) \longrightarrow 2CO$$

$$C + O_2\,(\text{excess}) \longrightarrow CO_2$$

$$S_8 + 8O_2 \longrightarrow 8SO_2$$

$$P_4 + 5O_2(g) \longrightarrow P_4O_{10}(s) \ldots\ldots\ldots \text{ excess oxygen}$$

$$P_4 + 3O_2(g) \longrightarrow P_4O_6(s) \ldots\ldots\ldots \text{ limited oxygen}$$

Uses of oxygen

- Oxygen is widely used as an oxidizing agent.
- Most of the oxygen produced is used in the production of steel.
- It is used in the production of metals
- With acetylene, oxygen is used in oxyacetylene welding.
- It is used to support life in medicine, in air and space travel, and in submarines.
- It is also used as bleaching agent for compounds and in biological treatment of waste water, etc

Illustrative Examples

39. Which of the following is used in the production of steel?

A. Nitrogen
B. Sulphur
C. oxygen
D. phosphorus

Explanation: Oxygen is widely used in the production of steel.

Answer C

40. Which of the following processes is used for the industrial production of oxygen? (EGSECE……..2007E.C)

A. Fractional distillation of liquid air.
B. Photosynthesis
C. Thermal decomposition of oxides
D. Electrolysis of water

Answer A

41. All of the following compounds can be used as raw material to manufacture oxygen in the laboratory. EXCEPT

A. $KClO_3$
B. HgO
C. Ag_2O
D. $CaCO_3$

Answer D

42. What are the common oxidation states of oxygen in compounds?

Solution

- -2 in most compounds
- -1 in peroxides
- -1/2 in superoxides
- +2 in OF_2

43. What are the two allotropes of oxygen?

Solution:

O_3 and O_2

44. What are the two gases manufactured industrial by the same method?

Solution:

O_2 and N_2 by fractional distillation of liquid air.

iv Sulphur

Occurrence

- Sulphur is found as the elemental form in nature.
- It is also found in combined form as:

A. Sulphides such as:

- Galena – PbS
- Cinnabar – HgS
- Pyrite (Fool's gold) – FeS_2
- Sphalerite or zin blend – ZnS

B. Sulphates such as:

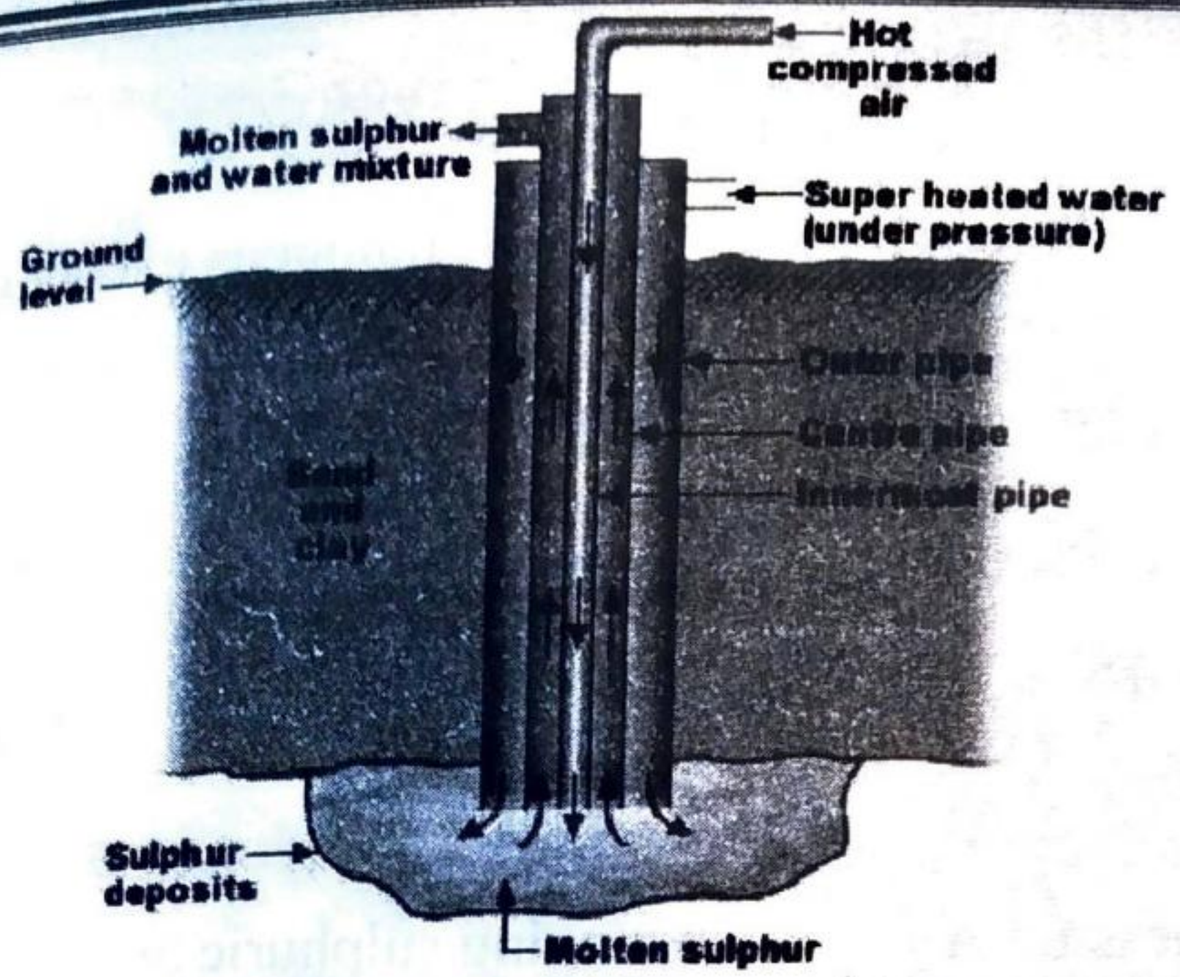
- Gypsum – $CaSO_4.2H_2O$
- Barite – $BaSO_4$

- Natural gas contain hydrogen sulphide

Production of sulphur

- Sulfur is extracted from underground deposits of elemental sulfur by the **Frach process**

- Super heated steam – melts the sulfur in the ground
- Compressed air – forces the froth of sulfur from the ground to up.



Frasch process for the extraction of sulphur (or mining of sulphur).

Figure 9.1 The Frash process

Physical properties of sulfur

- Sulphur has two important allotropes. These are **rhombic and monoclinic sulphur.**

Table 9.2. Different properties of mono clinic and Rhombic sulfur

Rhombic sulfur	Monoclinic sulfur
➦ Yellow, tasteless and oderless solid	➦ Consists of S_8 molecules
➦ Stable form of sulphur containing S_8 molecules	➦ Less stable form of sulphur
➦ Insoluble in water	➦ Insoluble in water but soluble in CS_2
➦ Soluble in CS_2	➦ Stable between 96^0c and 119^0c
➦ Converted to monoclinic sulphur when heated	
➦ Stable below 96^0c	

Chemical properties of sulfur

- Sulphur reacts with active metals

$$16K(s) + S_8(s) \longrightarrow 8K_2S(s)$$

- It reacts with non-metals

$$S_8(g) + 8O_2(s) \longrightarrow 8SO_2(g)$$

$$S_8(g) + 16F_2(g) \longrightarrow 8SF_4(g)$$

$$S_{8(g)} + 24F_2(g) \longrightarrow 8SF_6(g)$$

- In general, sulfur is oxidized by all of the halogens except **iodine which** is not sufficient oxidizing

$$S_8 + 16x_2 \longrightarrow 8Sx_4 \ (x=F, Cl)$$

$$S_8 + 8x_2 \longrightarrow 8Sx_2 \ (x=F, Cl)$$

$$S_8 + 4x_2 \longrightarrow 4S_2x_2 \ (x=F, Cl, Br)$$

Uses of sulphur

- Sulphur is used widely in making sulphuric acid.
- It is used in the **Vulcanization of rubber,**
- **Sulphur is used in the manufacture of insectides, fungicides and gunpowder.**
- **Sulphur has applications in the manufacture of skin ointments and drugs**
- **It is a good insultator**
- **It is a minor conituent of fats, body fluids and skeletal minerals.**

Illustrative Examples

45. Which of the following element is used to vulcanize rubber? (EGSECE……2002E.C)

A. Phosphorus
C. Carbon
B. Sulphur
D. Chlorine

Answer B

46. Which one of the following processes is used to obtain sulphur from the underground sulphur deposited?

A. Frasch process
C. Solvay process
B. Ostwald process
D. Contact process

Solution

Ostwald process – nitric acid.

Solvay process – sodium carbonate

Contact process – sulphuric acid

Frasch process – sulphur

Answer: A

47. What is the main purpose of sulphur?

Solution:

production of sulphuric acid

48. What are the two important allotropes of sulphur?

Solution:

Monoclinic and Rihombicsulphur.

v Chlorine

Occurrence

- Chlorine is found in nature in combined form as **chlorides of** sodium, potassium, calcium and magnesium.
- Sodium chloride is the main ore of chlorine

Preparation of chlorine

- Industrially, chlorine is manufactured by **electrolysis of concentrated sodium chloride solution /brine/.**

$$2NaCl(aq)+2H_2O(1) \rightarrow Cl_2(g) + 2NaOH(aq) + H_2(g)$$

- In the laboratory chlorine can be prepared by the **following** reaction:

$$MnO_2(s) + 4HCl(aq) \longrightarrow MnCl_2(g) + Cl_2(g) + 2H_2O(1)$$

Physical properties of chlorine

- It is greenish – yellow gas.
- It melts at -101°c and boils at -34°c.
- It is poisonous gas.

Chemical properties of chlorine

- It reacts with active metals

$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

$Zn + Cl_2 \longrightarrow ZnCl_2$

- It reacts with hydrogen

$Cl_2 + H_2 \longrightarrow 2HCl$

- Chlorine displaces bromine and iodine from their compounds.

$Cl_2(g) + 2NaBr(s) \longrightarrow Br_2(g) + 2NaCl(s)$

$Cl_2(g) + 2NaI(s) \longrightarrow I_2(g) + 2NaCl\ (s)$

- Chlorine reacts with water to form hydrochloric acid and hypochlorous acid. This reaction of chlorine is called **disproportionation reaction**

$Cl_2(s) + H_2O(1) \longrightarrow HCl(aq) + HClO\ (aq)$

Uses of chlorine

- Chlorine is a very strong oxidizing agent, which is used commercially as a bleaching agent and as a disinfectant.
- Chlorine is widely used for producing safe drinking water.
- It is also extensively used in the production of dyestuffs, textiles, petroleum products, medicines, antiseptics, insecticides, plastics.
- It is also used to make solvents such as carbontetra chloride (CCl_4), chloroform ($CHCl_3$), Dichloroethylene ($C_2H_2Cl_2$) and trichloroethylene (C_2HCl_3)

Illustrative Examples

49. Which one of the following chemicals is used to disinfect water? (EGSECE…..2003E.C)

A. Fluorine
B. Nitrogen
C. Oxygen
D. chlorine

Answer D

50. Which of the following is the most abundant halogen?

A. Cl_2
B. F_2
C. I_2
D. Br_2

Answer A

51. Why chlorine displaces only Bromine and iodine but not fluorine from their compounds?

Explanation

➦ More reactive element displaces less reactive element. Chlorine is more reactive than bromine and iodine but less reactive than fluorine.

52. Balance and complete the following reactions.

a. $F_2 + KCl \longrightarrow$
b. $Cl_2 + KBr \longrightarrow$
c. $Br_2 + KI \longrightarrow$
d. $Cl_2 + KI \longrightarrow$

Solution

a. $F_2 + 2KCl \longrightarrow 2KF + Cl_2$
b. $Cl_2 + 2K\,Br \longrightarrow 2KCl + Br_2$
c. $Br_2 + 2KI \longrightarrow 2KBr + I_2$
d. $Cl_2 + 2KI \longrightarrow 2KCl + I_2$

9.4 Some Industries in Ethiopia

Table 9.3. Some industries in Ethiopia

Industry	Product	Location
Nifas silk paint factory	Paint	Addis Ababa
Dil paint factory	Paint	Addis ababa
Tseday paint factory	Paint	Addis Ababa
Repi soft and detergent factor	Soap and detergent	Addis Ababa
Gulele soap factory	Soap	Addis Ababa
Matador Addis Goma factory	Tyres	Addis Ababa
Saint George Brewry	Beer	Addis Ababa

Awash Tanney	Processed leather	Addis Ababa
Addis Ababa foam and plastic factory	Plastics	Addis Ababa
Ethioplastic	Plastic	Addis Ababa
Ethiogas and earts	Plastic arts and CO_2	Addis Ababa
Chora gas and chemical products	Oxygen, acetylene, shoe polish, floor wax	Addis Ababa
Nas foods Factory	Biscuits	Addis Ababa
Addis Glass Factory	Biscuits	Addis Ababa
Sulphuric acid and aluminum sulphate factory	Sulphuric acid and aluminiumsulphate	Awash Mekasa
Tabor ceramics factorfy	Ceramics	Hawasa
ArbaminchTextiel factory	Textile	Arbaminch
Almeda textile factory	Textile	Adwa
Caustic soda factory	Coautic soda	Zeway
Adamituluplasticide factory	Pesticide	Adamitulu
Mojo Tannery	Processed leather	Mojo
Dashen Brewery	Beer	Gondar
Harar Brewery	Beer	Harar
Meta brewery	Beer	Sebeta
Bedelle brewery	Beer	Bedelle
Wonji sugar factory	Sugar	Wonji
Metehara sugar factory	Sugar	Metehara
Fincha sugar factory	Sugar	Fincha
Cement factories	Cement	Tigray, mugger, diredawa, northenshewa

1. Glass

Glass is amorphous solid, consists of two or more silicates.

Types of glass

1. **Soda – lime glass** – is a mixture of sodium silicate (Na_2SiO_3) and calcium silicate ($CaSiO_3$)
   - ➦ Used to make bottles, dishes, etc

2. **Borosilicate glass (pyrex glass)** – consists of boronoxide.
   - ➦ Resist chemical corrosion and temperature changes.
   - ➦ Used to make laboratory equipments such as test tubes, flasks and beakers.

3. **Quartz glass** – is formed from pure silica.
   - ➦ is highly transparent.

Colored glasses are obtained by the addition of metallic oxides. For example, cobalt(II) oxide produces blue glass, chromium (III) oxide produces orange glass, ferrous compounds produces green glass, cadmium sulphideyellow glass, cupricoxide produces green glass, sulphur produces amber glass

Steps in glass production

1. **Batch preparation** – preparing raw materials for the production of glass.
2. **Glass melting** – raw materials and recyled glasses are fed into a glass-melting furnace.
3. **Glass forming** – the melted glass is changed into required shape.
4. **Annealing** – removing internal stress by reheating the glass
5. **Inspection** – testing of the glass product.
6. **Packing and dispatching** – distribution.

2. Ceramics

- ➦ **Clay, talc and feldspar** are used as traditional ceramics.
- ➦ The raw material for industrial ceramics are: silicon, carbide, alumina, barium titanate, and titanium carbide.
- ➦ Ceramics are mainly used to make dishes, kitchen, tiles and bathroom structures. To lesser extent, used as insulators and as artificial bones.

➦ In the future, the ceramics may be used to remove impurities from drinking water and to substitute diseased heart valves.

3. Cement

Cement is made by heating limestone, alumina and silica bearing material such as clay in a **kiln** at very high temperature.

Port land cement is important building material. It is a mixture of **calcium and aluminum silicate.**

➦ Cement mainly contains:

- Calcium silicate – $CaSiO_3$
- Calcium aluminate – $CaAl_2O_4$
- Iron (III) oxide – Fe_2O_3
- Magnesium oxide – MgO

Manufacturing process

1. In the upper part of the kiln, raw materials are heated to remove moisture
2. In the middle part of the kiln, lime stone /$CaCO_3$/ is decomposed by heat.
3. In the lower part of the kiln, at high temperatures the following reactions take place.

$$2CaO + 2SiO_2 \longrightarrow 2CaO.SiO_2$$

$$3CaO + 3SiO_2 \longrightarrow 3CaO.SiO_2$$

$$3CaO + 3Al_2O_3 \longrightarrow 3CaO.Al_2O_3$$

$$4CaO + 4Al_2O_3 + 4Fe_2O_3 \longrightarrow 4CaO.Fe_2O_3$$

$$MgO + SiO_2 \longrightarrow MgSiO_3$$

➦ The end mixture of all these silicates and aluminates is called **cement clinker.**

➦ After cooling, it the cement clinker is mixted with gypsum and powdered.

➦ Cement mainly contains calcium oxide, iron (III) oxide and sulphutrioxide in decreasing order of percentage.

➦ Cement is mainly used for construction and buildings.

Illustrative Examples

53. Cement is made by heating (EGSECE……2008E.C)

A. Aluminum oxide and gypsum

B. Hematite, siderite and magnesiumoxide

C. Limestone, alumina and clay

D. Calcium oxide and silicon dioxide

***Explanation*:** Cement is made by heating limestone ($CaCO_3$), alumina (Al_2O_3) and silica bearing material such as clay at 1450°c in a kiln by a process called **calcinations.**

Answer C

54. Which of the following is NOT the use of ceramics? (EGSECE…… .2004E.C)

A. Bath room structures

B. Kitchen tiles

C. Clothing

D. Art sculptors

Answer C

55. The chief compositions of glass are:

A. Sodium chloride and calcium silicate

B. Sodium silicate and calcium silicate

C. Calcium silicate and calcium aluminate

D. Calcium carbonate and silicondioxide

Answer B

56. Compounds that occur widely in glass, cement and ceramics are (EGSECE……2002E.C)

A. Silicates

B. Sulphates

C. Carbonates

D. Photphates

Answer A

57. Cement consists of chiefly one of the following mixtures?

A. $Ca(OH)_2$ and $CaSiO_3$

B. Na_3AlF_6 and $Al_2O_3.2H_2O$

C. Na_2SiO_3 and $CaSiO_3$

D. $CaSiO_3$ and $Ca(AlO_2)_2$

Solution:

Cement mainly contains calcium silicate, calcium aluminate, iron (III) oxide, and magnesium oxide.

Answer: D

58. List the raw materials used to make ordinary glass.

Solution

Soda – lime glass is ordinary glass which is a mixture of sodium silicate and calcium silicate. It is made by heating a mixture of silica sand, sodium carbonate or sodium-sulphate and limestone.

59. Write the processes of manufacturing ceramics

Solution

- Moulding
- Densification

60. What are the most popular manufacturing industries in Ethiopia?

Solution:

Sugar factory, cement factory, ceramics and tannery

61. Identify the substances added to form the following colored glasses.

a. Deep blue glass
b. Green glass
c. Yellow glass
d. Orange glass
e. Amber glass

Solution

a. Cobalt (II) oxide
b. Iron (II) compounds and copper (II) oxide
c. Cadmium sulphide
d. Chromium (III) oxide
e. Sulfur

62. What type of glass is used to make laboratory of equipments such as test tubes, beakers and flasks?

Solution

Borosilicate glass or pyrex glass

63. Write three tradition ceramics

Solution

Porcelain, tiles and pottery, they are formed from minerals such as clay, talc and feldspare.

64. Arrange the following components of cement increasing order of their percentage.

CaO, Fe_2O_3, SiO_2, Al_2O_3, SO_3, MgO

Solution

$SO_3 < Fe_2O_3 < MgO < Al_2O_3 < SiO_2 < CaO$

4. Sugar

➦ Raw materials = sugar cane and sugar beet
steps in sugar production.

A. Collecting the harvest

➦ Cutting the sugar cane by removing leaves and transporting to the sugar factory.

B. Cleansing and Grinding

➦ Cleaning the sugar cane, chopping the cane in to pieces.

C. Juicing

➦ Extracting juice and clarifying it.

D. Clarifying

➦ Using carbondioxide and milk of lime purifying.

E. Evaporation

➦ The clarified juice evaporated into thick brown syrup.

F. Crystallization

➦ Forming crystals of sugar by removing the extra water.

G. Refinery

➦ Raw sugar is transported to a cane sugar refinery for the removal of molasses, minerals and other non-sugars that contaminate it.

➦ The brown syrup is bleached by **sulphurdioxide (SO_2)**

H. Separating and packaging

- Large and small crystal of sugar, packed and labeled as white pure sugar.

5. Paper and Pulp

- Raw material = wood
- Wood pulp is dry fibrous material used for making paper.
- Pulp wood is the timber resources used to make wood pulp.
- **Spruce, pine, fir, larch and hemlock, eucalyptus, aspen and birch** are used to make wood pulp. The first five are soft woods and the last three are hard woods.

Steps in making pulp and paper

1. Harvesting

- Cutting trees and transporting into the pulp and paper factory.

2. Preparing for pulping

- Bark of tree removed and the wood is chipped and screened

3. Pulping

- Making wood pulp.
- This can be done by two processes.

A. Mechanical pulping

- Uses steam , pressure and temperature without chemical to tear fibers.

Disadvantage

The quality of fiber decreases due to short, weak fibers obtained by mechanical pulping.

B. Chemical pulping

- Uses chemicals to tear fibers.
- It is divided into two.

i The kraft process

- It is the main chemical pulping method

- It is widely used method.
- It uses aqueus sodium hydroxide and sodium sulphide as a digestion solution.
- It uses basic digestion medium.

ii The sulphate process

- It uses sodium or magnesium bisulphate digester.

4. Bleaching

- The bleaching agents in pulp and paper factory are **chlorine, chlorine oxide, ozone and hydrogen peroxide.**

5. Making paper from pulp
   - Producing paper

6. Tannery

Tanning is a process of changing animal skin into leather using tannin.

Tannin is an acidic chemical that change protein structure of skin permanently.

Steps in leather production

1. Preparation stages

Preparing the skin for tanning.

2. Curing

Salting or drying the skin by using brine.

3. Soaking

Placing the hides in water for several hours.

4. Flesh remove

Removing flesh from hides using sharp machines or knives.

5. Hair removal

- The hair removed from hides using lime and water.

6. Scudding

- Removing any left hair with machines

7. Deliming

➦ Neutralizing the lime using vat of acid.

8. Tanning

➦ Converting the protein of raw skin to stable material. This can be done using.

i Natural tanning or vegetable tanning

➦ This method uses the bark and leaves of many plants as tannin.

ii Mineral tanning

➦ This method uses chemicals such as chromium sulphate and other salts of chromium as tannin.

➦ It is faster than natural tanning.

9. Crusting

➦ It is the last step and includes coloring and rolling the leather.

7. Food processing and preservation

There are two methods used to preserve food.

A. Traditional methods

These include the following

1. Salting – removes moisture
2. Pickling – preserving food in an edible liquid that inhibits the growth of micro-organisms.
3. Sugaring – sugar is used to preserve fruit such as apples, pears and peackls.
4. Smoking – preserving food by exposing to smoke.
5. Drying – used to preserve meat by removing water from it.
6. Canning – sealing food in sterile cans or jars and boiling the contents to kill or weaken any remaining bacteria in the form of sterilization.

B. Modern methods of food preservation

i Freezing

➦ Used to preserve wide range of foods

ii Vacuum – packing

➦ Stores food in a vacuum environment most of the times in an air – tight bag bottle

Illustrative Examples

65. Which of the following steps are correct in the preparatory stages of tanning? (EGSECE…..2008E.C)

A. Deliming⟶ curing ⟶ hair removal ⟶ scudding ⟶ soaking ⟶ flesh removal

B. Curing ⟶ socking ⟶ flesh removal ⟶ hair removal ⟶ scudding ⟶deliming

C. Soaking ⟶deliming⟶ curing ⟶ flash removal ⟶ scudding ⟶ hair removal

D. Hair removal ⟶ flash removal ⟶ curing ⟶ soaking ⟶ scudding ⟶deliming.

***Explanation*:** The correct order in the preparatory stages of tanning is

Curing ⟶ soaking ⟶ flesh removal ⟶ hair removal ⟶ scudding ⟶deliming.

Answer B

66. The purpose of freezing in food preservation is to

A. Suppress the growth of bacteria and micro-organisms

B. Keep it from the access of house hold pests

C. Increase air circulation in food substance

D. Improve the taste and odour of the food item

Answer: A

67. How does protein substance in animal hide or skin be prevented from bacterial infection during tanning?

A. By curing using brine

B. By scudding using plastic tool

C. By soaking in water

D. By deliming using an acid

Explanation

➦ Brine curing is the simplest and fastest method to prevent protein substance, collogen from bacterial infection.

Answer: A

68. In the tanning process, the purpose of putting the skin into slacked lime solution is to

A. Facilitate the removal of unwanted flesh

B. Dehydrate the skin

C. Facilitate the removal of hair

D. Soften the skin

Answer C

69. What substance is used to bleach the brown syrup in the process of sugar manufacturing?

A. Potassium chlorate

B. Sulfur dioxide

C. Calcium chloride

D. sodium hydrochlorite.

Explanation: Bleaching is the process of removing color. This is done by using sulphurdioxide in sugar factory.

Answer B

70. Food can be preserved by one of the following methods. Which one is it?

A. Applying an oxidizing agent

B. Providing heat and oxygen

C. Keeping wet and moist

D. Storing in an air-tight bag or bottle.

Answer D

71. The purpose of adding salt to the skin in the process of tanning is

A. Make the skin soft

B. Dehydrate the skin

C. Remove the unwanted flesh from the skin

D. Loosen the outer layer of the skin

Explanation: Adding salt to the animal skin kill bacteria and dehydrate it.

Answer B

72. List the bleaching agents in pulp and paper factory.

Solution

- Hydrogenperoxide (H_2O_2)
- Chlorine oxide (Cl_2O)
- Chlorine (Cl_2)
- Ozone (O_3)

73. List the traditional methods used to preserve food.

Solution:

Salting, pickling, sugaring, smoking, drying and canning.

74. What is food preservation?

Solution:

Food preservation is the process of treating and handling food to stop or greatly reduce spoilage, loss of quality, edibility or nutritive value caused or accelerated by micro-organisms.

9.5 Environmental Pollution

Pollution is the discharge of harmful waste materials to the environment.

Materials that cause pollution are called **pollutants.**

1. Air pollution

- It is mainly due to combustion reactions.
- The common air pollutants are
  - Sulphur dioxide
  - Oxides of nitrogen
  - Hydro carbons
  - Carbodioxide
  - Carbon monoxide
  - Particulates
  - Chloroflour carbons (CFC's)
  - Heavy metals /lead compounds/

- Sulphurdioxide and oxides of nitrogen cause **acid rain.**
- Hydrocarbons are responsible for photochemical smog.
- Carbonmonoxide when inhealed reduces the oxygen-carrying capacity of blood by reacting with haemoglobin.
- Carbondioxide increase the temperature of an environment.
- Particulates affect the breathing system
- Chlorofloro carbons deplete the ozone layer
- Heavy metals – damage the brain, kindneys and liver if the amount of lead is at high level.

Methods of reducing air pollution

- **From cars** can be fitted with exhaust systems containing special catalysts that make the harmful gases react together to produce safe gases.
- **From homes:** People should use smokeless fuels on their fins.
- **From factories and power stations:** use the methods below:

i. **Tall chimneys** – carry smoke, dust and waste gases high up into the air. But this is not good solution.

ii. **Scrubbers** – are tanks where the waste gases sprayed by jets of water before it reaches the chimneys.

iii. **Electrolytic precipitors** – the waste is led into a chamber called an electrolyte precipitator, where an electric current is passed through it. The smoke and dust particulates get charged and cling to the electrodes. In time, they fall to the botton of the precipitator and are removed.

- Reducing carbondioxide emission

2. Water pollution

- Every day, hundreds of harmful substances find their way into our rivers, oceans, lakes and seas. This causes a water pollution.

The water pollutants include

- Fertilizers
- Untreated sewage
- Detergents
- Waste from factories
- heat
- oil

Methods of reducing water pollution

- Treating wastes before discharging into water bodies.
- Industries should not discharge hot water that heats tanks and pipes in to the river.
- Recycling wastes
- Using natural fertilizers and biological method of controlling pests.

3. Land Pollution

- It is caused by the discharge of wastes on it.
- Land pollutants include:
  - Dumping of non-biodegradable substances such as plastics.
  - Leaching of harmful chemicals and metal ions.
  - Excessive use of synthetic fertilizers and chemicals.
  - Spillage of oil from leaking pipe lines.

Methods of reducing land pollution

- Manufacturing biodegradable plastics
- Recycling wastes
- Using natural fertilizers

Illustrative Examples

75. One of the following methods is used to reduce air-pollution. Which one is it? (EGECE....2008E.C)
    A. Recycling of agricultural wastes
    B. Recycling of non-biodegradable materials
    C. Increasing the use of organic fertilizers
    D. Reducing carbondioxide emission.

    Answer D

76. What is the direct effect of nitrates and phosphates as water pollution? They
    A. Cause the death of water organisms
    B. Increase the amount of insecticides or harbicides.
    C. Allows the spread of water born disease such as cholera or typhoid.
    D. Accelerate the growth of surface-water plants in a lake or river.

    Answer D

77. Which of the following industrial activities has a minimum potential in polluting the environment?
    A. Making use of petroleum fuel in running industries
    B. Applying phosphate and nitrate fertilizers to maximize crop production
    C. Applying insecticides and herbicides in forming
    D. Making use of hydroelectric power to generate electricity

Answer D

78. Which of the following methods cannot be used to reduce water pollutions?
    A. Recycling agricultural and industrial wastes
    B. Discharging heat /adened water into natural water systems
    C. Treatment of water before discharge into rivers and lakes
    D. Using moderate amount of agricultural chemleau.

Answer B

79. Among the following pollutants, which pair causes photo chemical smog?
    A. Particulates and lead compound
    B. Nitrogen and sulphur compounds
    C. Carbon monoxide and ozone
    D. Hydro carbons and ozone.

Explanation

- Particulates and lead compound affect breathing system.
- Nitrogen and sulphur oxides cause acid ain

Answer D

80. The substances that deplete the ozone layer and increase the global temperature are_____.

Solution

- Chloroflouro carbons deplete the ozone layer as follows.

$$CF_2Cl_2 \rightarrow CF_2Cl\bullet + Cl\bullet$$

$$Cl\bullet + O_3 \longrightarrow ClO\bullet + O_2$$

Solved problems

81. What are the major constituents of acid rain?

Solution:

Carbondioxide (CO_2) and sulphurdioxide (SO_2)

82. Eutrophication can be caused by an in appropriate use of _____

Solution:

Phosphate and nitrate fertilizers.

83. Write the effects of acid rain.

Solution:

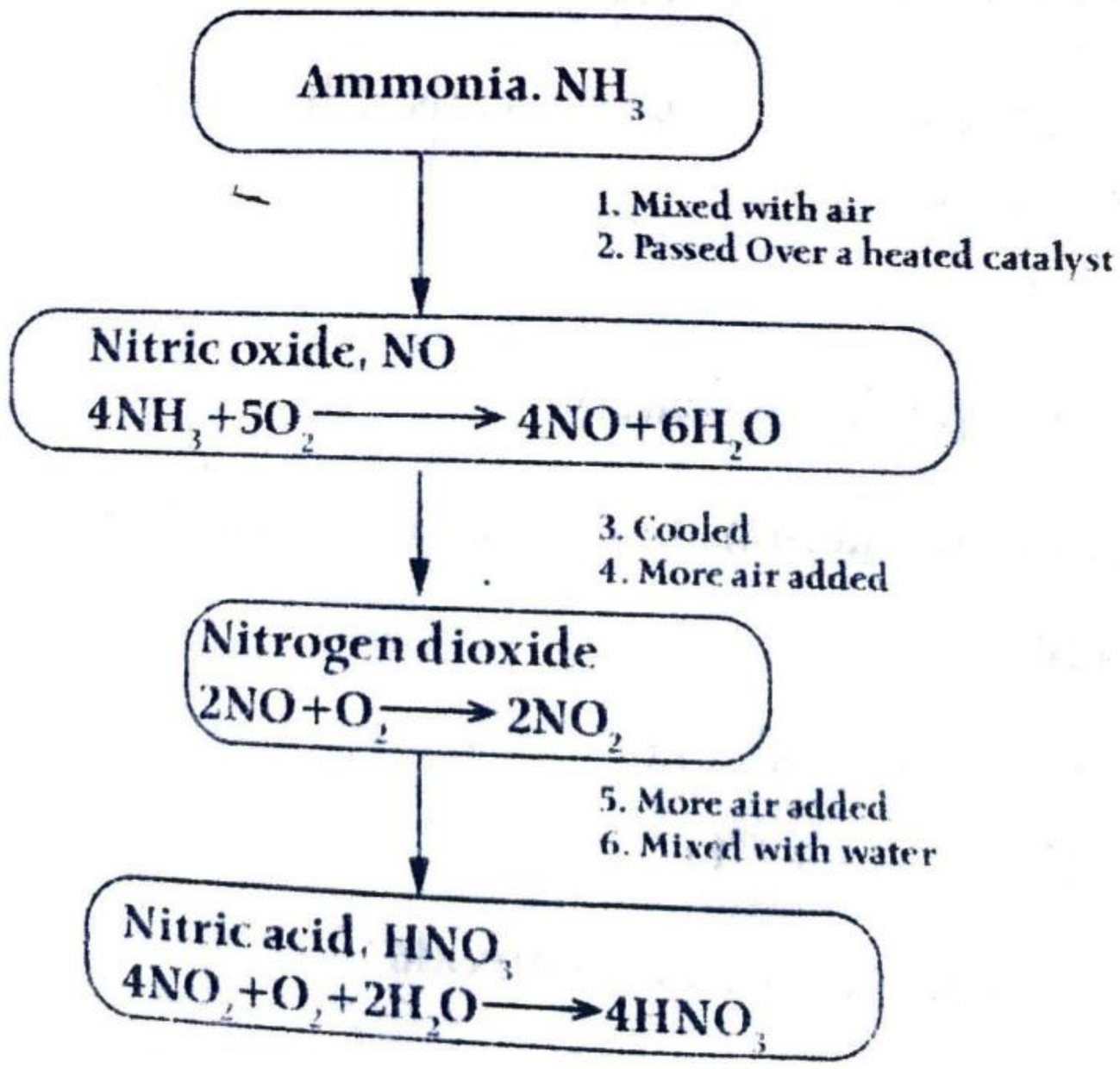
- Acid rain accelerates corrosion of metals, destroy marble building, drop the PH of water and dissolves the green pigments of plants.

84. What are the three important methods that converts pig iron into steel?

Solution

- The Bessemer converter
- The open heart furnance
- The basic oxygen convertor

85. Use flow chart to show the stages used to make nitric acid by Ostwald process.



86. Write the four steps involved in the production of sulphuric acid by contact process.

Solution

Step – 1 $S + O_2 \longrightarrow SO_2$

Step – 2 $2SO_2 + O_2 \xrightarrow{V_2O_5(s)} 2SO_3$

Step – 3 $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$

Step – 4 $H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$

87. Classify the following as heavy or light metals.

a. Au c. Al e. K g. Ag

b. Fe d. Cu f. Mg h. Ca

Solution

➦ a, b, d and g are heavy metals

➦ c, e,. f and h are light metals

88. Chlorine dissolves in water and reacts with it to form acidic solution. Write the balanced chemical equation.

Solution

$Cl_2 + H_2O \longrightarrow HClO + HCl$

Hyphochlorosis acid - HClO

89. Mention three diseases that are caused by drinking water polluted by untreated sewage.

Solution

Cholera, typhoid, dysentery

90. Write the main reactions that occur in the blast furnace.

Solution

iii Oxidation of coke

$C + O_2 \longrightarrow CO_2$

iv Reduction of carbondioxide

$$CO_2 + C \longrightarrow 2CO$$

v Reduction of iron oxides to iron

$$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$$

$$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$$

$$FeO + CO \longrightarrow Fe + CO_2$$

vi Decomposition of lime stone

$$CaCO_3 \longrightarrow CaO + CO_2$$

vii Reaction of quick lime with impurities such as silica

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

91. Plants do not take atmospheric nitrogen directly from the atmosphere. Why?

Solution

The triple bond in nitrogen (N_2) cannot be broken easily.

92. What are the main sources of nitrogen, oxygen, phosphorus, sulphur and chlorine?

Solution

Nitrogen and oxygen – atmospheric air

Phosphorus – calcium phosphate

Sulphur – underground deposit

Chlorine – sodium chloride

93. Write the approximate formula of rust and greenish layer on copper.

Solution:

Rust = $Fe_2O_3 . xH_2O$

Greenish layer = $Cu(OH)_2 . CuCO_3$

94. Give two reasons why aluminum is used to make cables?

Explanation

It is ductile, corrosion resistance, light and good conductor of electricity.

95. Arrange the following metals in increasing order of conductivity. Ag, Au and Cu

Solution:

$Au < Cu < Ag$

96. What is the formula of laughing gas? Explain why nitrous oxide is laughing gas.

Solution:

N_2O , because it gives a sense of laugh when inhaled.

97. What is the catalyst used in the contact process?

Solution:

Vanadium (v) oxide, V_2O_5

98. What is the main purpose of ammonia?

Solution: To make fertilizer

99. Name the waste gases released from the blast furnace.

Solution

Cabon monoxide, carbondioxide and nitrogen dioxide.

100. What is the gas that relights a glowing splint?

Solution: Oxygen (O_2)

Supplementary Exercise on Unit Nine

I. Match compounds listed on the left with their common name listed on the right

A	B
1. Al_2O_3	A. Quick lime
2. Na_3AlF_6 .	B. Sphalerite
3. Fe_2O_3	C. Galena
4. Fe_3O_4	D. Cinnabar
5. $FeCO_3$	E. Tenorite

6. FeS_2	F. Cuprite
7. $CuFeS_2$	G. Covellite
8. Cu_2S	H. Chalcocite
9. CuS	I. Chalcopyrite
10. Cu_2O	J. Fool's gold
11. CuO	K. Siderite
12. HgS	L. Magnetite
13. PbS	M. Hematite
14. ZnS	N. Cryolitte
15. CaO	O. Corundum
	P. Slaked lime

II. Choose the best answer from the given alternatives.

16. Which one of the following is not an ore of iron?

A. Hematite C. Siderite

B. Magnetite D. Chalcopyrite

17. The second abundant element in the earth's crust is

A. Silicon C. Iron

B. Oxygen D. Aluminum

18. Which of the following is not a use of nitrogen

A. Making matches C. production of ammonia

B. Making fertilizers D. as a refrigerant

19. Sulphuric acid is prepared commercially in four steps of reactions by the contact process

I $2SO_2 + O_2 \longrightarrow 2SO_3$

II $S + O_2 \longrightarrow SO_2$

III $H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$

IV $2SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$

Which of the following is the right sequence of the steps?

A. I, II, III and IV C. II, I, IV and III

B. II, III, I and IV D. I, II, IV and II

20. Which of the following is correct about the allotropes of phosphorus?
    A. White phosphorus is a waxy and reactive solid
    B. White phosphorus is soluble in water
    C. White phosphorus is more stable than red phosphorus
    D. The allotropes of phosphorus are not inter convertible
21. Which one of the following gases is a constituent of industrially useful fuel gases?
    A. Ammonia
    B. Oxygen
    C. Hydrogen sulphide
    D. Carbon monoxide
22. A mixture of sodium and calcium silicate is known as
    A. Glass
    B. Cement
    C. Ceramic
    D. Sand
23. Which of the following oxides of nitrogen is formed when lightning discharge passes through the air?
    A. NO B. NO_2 C. N_2O D. N_2O_3
24. The catalyst used in Haber process is
    A. V_2O_5 B. MnO_2 C. NO D. Fe
25. Which of the following reaction takes place in the manufacture of aluminum by Hall process?
    A. Aluminum oxides is decomposed by heat
    B. Cryolite is electrolyzed at 2000^0c
    C. Bauxite is electrolyzed using graphite electrode
    D. The impurities in the aluminum are removed by limestone
26. Which of the following is used in the manufacture of phosgene and mustard gas?
    A. N_2O
    B. CO
    C. NO_2
    D. Cl_2
27. Which of the following elements is stored under water to prevent it from reacting with oxygen?
    A. Magnesium
    B. Sodium
    C. White phosphorus
    D. rhombic sulfur

28. One of the following reactions is a confirmatory test for carbon dioxide. Which one is it?

A. $2Mg + CO_2 \longrightarrow 2MgO + C$

B. $C + CO_2 \longrightarrow 2CO$

C. $2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$

D. $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

29. The dense oily intermidate compound is the preparation of sulfuric acid has the formula

A. SO_2 C. $H_2S_2O_7$

B. SO_3 D. H_2SO_4

30. Which of the following metals is extracted from the ore known as bauxite?

A. Ca B. Mg C. Fe D. Al

31. What is the purpose of a converter in steel production?

A. To reduce the iron in the ore to elemental iron

B. To allow the addition of nitrogen for increase strength

C. To allow the slow solidification of the molten metal so that it will purify as it crystallizes

D. To remove impurity elements by oxidation

32. Which of the following does not contain sulfur?

A. Dolomite C. Galena

B. Iron pyrite D. Cinnabar

33. What is the major source of sulfur?

A. Sea water

B. Sulfides

C. sulfate

D. Underground deposits of element sulfur

34. The most abundant gas in the atmosphere is

A. H_2 B. N_2 C. O_2 D. CO_2

35. The most abundant noble gas in the atmosphere is

A. He
C. Ne
B. Ar
D. Kr

36. What are the three most abundant elements on earth?

A. Fe, Ca and Al
C. O, Si and Al
B. O, S and Ca
D. Si, M and O

37. Which of the following pairs are not allotropes?

A. Rhombic and monoclinic
B. Pig iron and stainless steel
C. Ozone and oxygen
D. White phosphorus and red phosphorus

38. What is the main commercial source of chlorine?

A. NaCl
C. $CaCl_2$
B. HCl
D. KCl

39. Which of the following is the purpose of treating the crude bauxite ore with hot aqueous sodium hydroxide in the extraction of aluminum by Hall process?

A. To get pure aluminum oxide
B. To decrease the solubility of aluminum oxide
C. To dissolve impurities like iron oxides
D. To lower the melting point of crude bauxite

40. Which of the following is renewable natural resource?

A. Coal
C. petroleum
B. natural gas
D. water

III. Give short answer

41. List the modern methods of food preservations.
42. State the two main types of tanning
43. In the hardening process of cement, the transition from plastic to solid state is _____

44. What are the chief raw materials used in the manufacture of Portland cement?

45. Complete and balance the following chemical equations

a. $P_4 + O_2 \longrightarrow$

b. $S_8 + O_2 \longrightarrow$

c. $Mg + N_2 \longrightarrow$

d. $Li + N_2 \longrightarrow$

e. $Fe + N_2 \longrightarrow$

Answer for Supplementary Exerciseon Unit One

Part I

1. True	3. True	5. False	7. False	9. True
2. False	4. False	6. True	8. True	10. True

Part II.

11. D	12. E	13. C	14. A	15. B

Part III.

16. A	23. A	30. A	37. B	44. B
17. D	24. C	31. A	38. D	45. A
18. C	25. A	32. B	39. C	46. D
19. D	26. B	33. D	40. A	47. A
20. A	27. B	34. B	41. B	48. A
21. B	28. C	35. B	42. A	
22. B	29. D	36. C	43. D	

Part IV

	Electrons	Protons	neutrons
49. a.	78	78	119
b.	40	42	55
c.	36	33	42

50. a. [Ar] $4s^2\ 3d^{10}\ 4p^5$
    b. [kr] $5s^2\ 4d^{10}$

51. a. 2, 8, 5
    b. 2, 8, 8

52. 80.5amu

53. 121.9 amu

54. 62.929

Answer to Supplementary Exercise on Unit Two

Part I.

1. True	3. False	5. False	7. False	9. False
2. True	4. True	6. False	8. True	10. True

Part II

11. D	13. H	15. G	17. B
12. C	14. E	16. F	18. A

Part III

19. C	23. B	27. A	31. B	35. A	39. B	43. A
20. D	24. D	28. C	32. A	36. B	40. C	44. A
21. C	25. D	29. B	33. D	37. B	41. D	45. A
22. A	26. C	30. C	34. D	38. B	42. D	

Part IV

46.

- ➦ atomic radius
- ➦ stability of electronic configuration
- ➦ charge carried by the anion

47.

a. F	c. Br	e. Be
b. Ca	d. Fe^{2+}	f. O^{2-}

48.

a. Ca b. F c. Mg^{+} d. Cr^{3+}

49.

a. Pb<Sn<Ge< As

b. I< Br <Cl< F

c. Cl< N < O < F

d. Li < Be < B < C

50.

a. period = 3

Group = VIIIA

b. period = 4

Group = VB

c. period = 5

Group = VA

Answer to Supplementary Exercise on Unit Three

Part I.

1. True
2. False
3. True
4. True
5. False
6. True
7. True
8. False
9. True
10. False

Part II.

11. C
12. B
13. D
14. C
15. A
16. B
17. C
18. A
19. B
20. C
21. C
22. D
23. A
24. B
25. C
26. A
27. A
28. D
29. B
30. D
31. B
32. A
33. C
34. D
35. B
36. A
37. D
38. A
39. D
40. C

Part III

41. A—D < A—C < A—B < A—E

42. a. $CH_3 CH_2OH$

    b. $CH_3 COOH$

    c. HF

43. London force < dipole – dipole force < hydrogen bond < ionic bond

44. $N_2 < Br_2 < K < Mg$

45. dipole – dipole force

    Hydrogen bond

    London force

Answer to Supplementary Exercise on Unit Four

Part I

1. D	5. D	9. A	13. C	17 A	21. A	25. A
2. B	6. D	10. C	14. A	18. B	22. C	26. C
3. C	7. B	11. D	15. C	19. B	23. B	27. A
4. B	8. C	12. B	16. A	20. D	24. C	

Part II

28. 207.5g/mol

29. 8×10^{22} atoms of H.

30.

a. C_6H_6 b. 8.57g of Br_2 c. 69.5%

31.

a. III c. IV e. endothermic

b. I d. II

32.

a. Iron + sulfuric acid → iron (II) sulphate + Hydrogen

b. phosphors + Oxygen → phosphors pent oxide

33.

- production of new substances
- generate electricity (exothermic reactions)

34.

a. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

b. $Na_2CO_3 + H_2S \rightarrow Na_2S + CO_2 + H_2O$

35. 62.16g

36. Homogeneous equilibrium is a chemical equilibrium in which products and reactants are in the same phase

Examples: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

$HF(aq) \rightleftharpoons H(aq) + F^-(aq)$

Heterogeneous equilibrium is equilibrium in which reactant and products are in different phases.

Example: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$CO(g) + Fe_2O_4(s) \rightleftharpoons CO_2(g) + 3FeO(s)$

37. Physical properties = a, c, f, g, h and J

Chemical properties = b, d, e and i

Answer to Supplementary Exercise on Unit Five

Part I

1. B	3. D	5. F	7. C
2. A	4. E	6. G	8. H

Part II

9. B	13. B	17. A	21. A	25. B	29. A
10. A	14. A	18. A	22. B	26. D	30. A
11. C	15. B	19. D	23. C	27. A	
12. B	16. C	20. C	24. A	28. B	

Part II

31. a) 3.95 atm b) 0.5atm c) 1atm

32. $v_2 = 5.46L$

33. $V_1 = 4.32L$

34. $P_2 = 591torr$

35. $V_2 = 280mL$

36. $m = 4.6g$

37. $V = 2.8L$

38. $v_2 = 405mL$

39. 72g/mol

40. molar volume

Answer to Supplementary Exercise on Unit Six

Part I

1. D	8. D	15. A	22. B	29. D	36. C
2. C	9. D	16. D	23. D	30. D	37. D
3. .B	10. B	17. B	24. D	31. C	38. A
4. C	11. A	18. C	25. D	32. B	39. C
5. C	12. A	19. A	26. C	33. B	40. D
6. C	13. A	20. C	27. A	34. A	41. B
7. A	14. B	21. B	28. B	35. C	

Part II

42. a 2 – bromo – 3 – chloro – 4 - ethyl - 2, 3 - dimethyl hexane

b. 4 - methyl - 3 - heptene

c. 4,4 – dimethyl – 2 – pentyne or

4,4 – dimethyl pent – 2 – yne

d. 4 – hexen – 2 – yne or Hex – 4 – ene – 2 – yne

e. 4 – methyl pentane – 2, 3 – diol

f. 3,5 – dimethyl – 5 – octen – 4 – 01

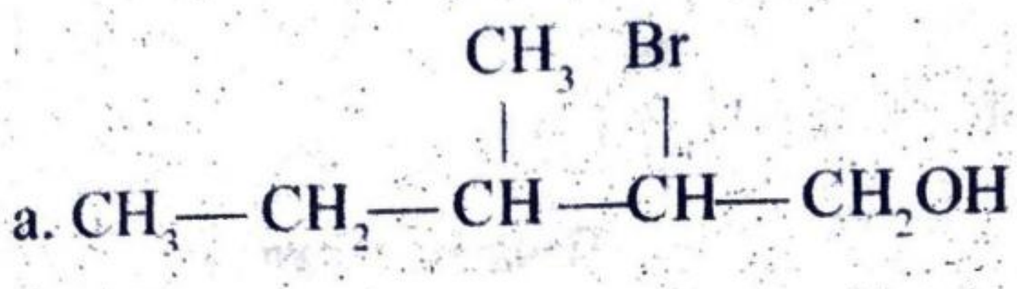
g. 2 – methyl butanal

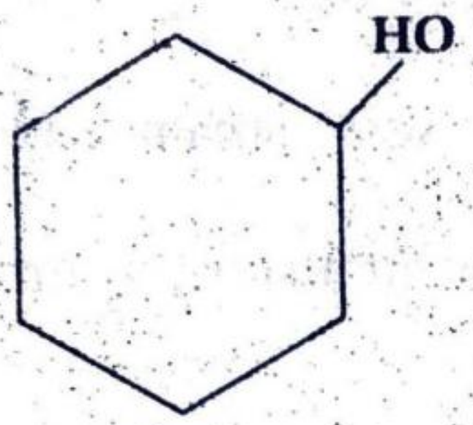
h. 3,3 – dimethyl butan - 2 - one

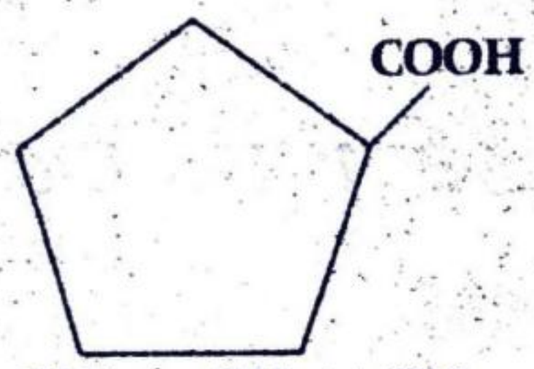
i. octanoic acid

j. n – butyl ethanoate or n – butyl acetate

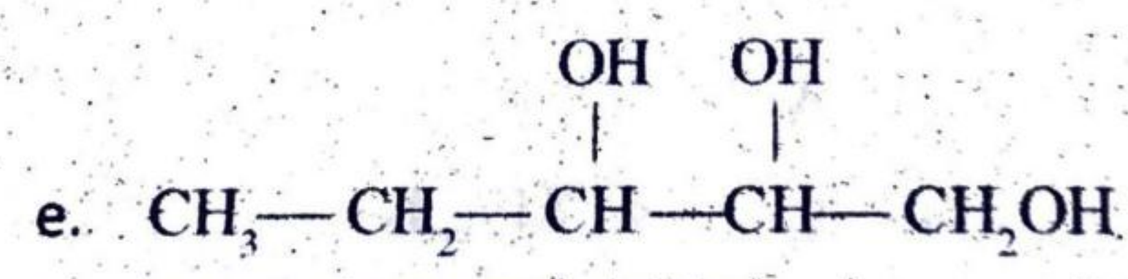
k. tert – butyl methyl ether or
2 – methoxy – 2 – methyl propane

a. 

b. 

c. 

d. $CH_3—CH_2—CH_2—C\equiv C—CH_2—CHO$

e. 

f. $HOOC\,CH_2CH_2\,COOH$

43. a. $\longrightarrow CH_3—CH_2—CH_3$

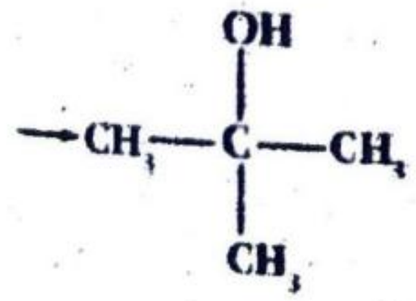
b. $\longrightarrow CH_3—CH_2—CH_3 + Na_2CO_3$

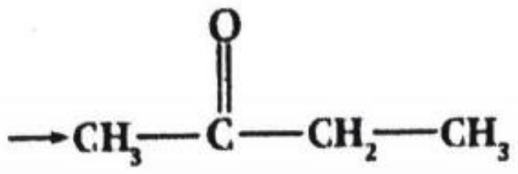
c. $\longrightarrow CH_3—C\equiv CH + 2KBr + 2H_2O$

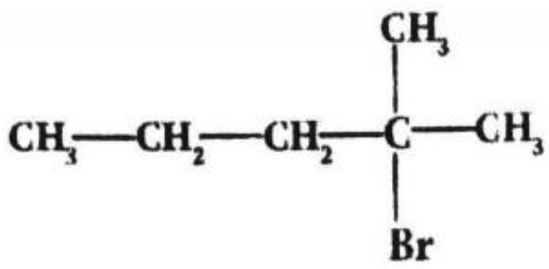
d. $\longrightarrow CH_2=CH—CH_3 + H_2O$

e. $\longrightarrow CH_3CH_2OH + KCl$

f. $\longrightarrow CH_3CH_2COOK + CH_3OH$

g. 

h. 

i. 

j. $\longrightarrow CH_3CH_2CH_2OOCCH_2CH_3 + H_2O$

44.

Advantages	**Disadvantages**
Protect crops from pests	pollutes water
	Causes soil contamination
	Poisons human and animal population

45. Ether

Answer to Supplementary Exercise on Unit Seven

Part I

1. C	6. D	11. A	16. C	21. C
2. D	7. B	12. C	17. B	22. C
3. B	8. D	13. C	18. D	23. B
4. B	9. C	14. B	19. A	24. B
5. A	10. B	15. A	20. A	25. A

Part II

26. 950mL

27. a, c, d and f are macro nutrients

    b and e are micro nutrients

28. a. Calcium phosphate

    b. Iron (III) carbonate or ferric carbonate

    c. sodium nitrite

d. Copper (I) oxide or Cuprous oxide

e. Manganese (IV) oxide

f. Potassium sulphite

g. Magnesium nitride

29. a. ⟶ salt + hydrogen gas

b. ⟶salt + carbon dioxide + water

c.⟶salt + sulfur dioxide + water

d. ⟶salt + water

30. pH scale is a scale between 0 and 14

0 – strongest acid 14 - strongest base

31. a. ORS is a mixture of in gradients such as common salt, sugar and water used as a hydrating beverage.

b. Hygroscopic salt is a salt which absorb water from the atmosphere but remain solid. $CuSO_4$

c. Deliquescent salt absorbs water from the atmosphere to form of **solution.**

Ex. $CaCl_2$, $NaNO_3$, $FeCl_3$, $MgCl_2$, $NaNO_3$

d. Efflorescent salts that lose their water of crystallization to the atmosphere.

Example: $NaCO_3$, $10H_2O$, $Na_2SO_4.10H_2O$

32. Used in fireworks.

33. Organic fertilizers – examples animals dung and decayed plant substances.

Synthesized fertilizer-examples $(NH_4)_2SO_4$,KNO_3, $(NH_4)_2HPO_4$, NH_4NO_3

34. $2KNO_3 \xrightarrow{heat} 2KNO_2 + O_2$

35. a. CO

b. NH_3 c. O^{2-}

Answer to Supplementary Exercise on Unit Eight

Part I

1. B	6. A	11. D	16. C	21. D
2. C	7. C	12. A	17. C	22. A
3. B	8. A	13. D	18. B	23. C
4. C	9. B	14. C	19. A	24. D
5. D	10. C	15. D	20. D	25. A

Part II

26. a. cation is positive ion while anion is negative ion.

    b. at anode oxidation occurs but at cathode reduction occurs.

    c. inert electrode does not react with electrolyte but active electrode reacts with electrolyte or product of electrolysis.

    d. strong electrolytes produce more ions.

27. a,b and c are not-electrolyte d, e and f are electrolytes.

28. non-electrolyte

29. electrodes

30. anode reaction

    $Zn \longrightarrow Zn^{2+} + 2e$

    Cathode reaction

    $Cu^{2+} + 2e \longrightarrow Cu$

Answer to Supplementary Exercise on Unit Nine

Part I

1.	O	3.	M	5.	K	7.	I	9.	G	11.	E	13.	C	15.	A
2.	N	4.	L	6.	J	8.	H	10.	F	12.	D	14.	B	16.	D

17. A	20. A	23. A	26. D	29. C	32. A	35. B	38. A
18. A	21. D	24. D	27. C	30. D	33. D	36. C	39. A
19. C	22. A	25. C	28. D	31. D	34. B	37. B	40. D

Part III

41. Freezing – Freeze – drying - vacuum – packing

42. vegetable or natural tanning - Mineral tanning

43. setting of cement

44. Limestone and clay

45. a. $P_4 + 5O_2 \longrightarrow P_4O_{10}$

b. $S_8 + 8O_2 \longrightarrow 8SO_2$ c. $3Mg + N_2 \longrightarrow Mg_3N_2$

d. $6Li + N_2 \longrightarrow 2Li_3N$ e. no reaction

Glossary

Acid: *A compound that ionizes in water to give hydronium ion and a negative ion.*

Activation energy: *The minimum amount of energy required for molecules to react.*

Actual yield: *The amount of product obtained experimentally.*

Alkane: *A hydrocarbon with all carbon - carbon single bonds.*

Alkene: *A hydrocarbon that contains at least one carbon carbon double bond.*

Alkyne: *A hydrocarbon that contains carbon - carbon triple bond.*

Alcohol: *An organic compound containing hydroxyl group as a functional group.*

Allotropes: *Different forms of the same element that exist in the same physical state but different structure.*

Alloy: *A homogeneous mixture of two or more than two metals.*

Anion: *Negative ion*

Anode: *The electrode where oxidation occurs in electrochemical cell.*

Atom: *The smallest particle of an element that retains the property of that element.*

Atomic number: *The number of protons in the nucleus of an atom.*

Avogadro's law: *The volume of a gas si directly proportiona to the quantity of the gas at the same temperature and pressure.*

Avogadro's number: *The number of particles in a mole of any substance.*

Battery: *Store chemical energy and convert it into electrical energy.*

Boyle's law: *The volume of a gas is inversely proprotional to pressure at constant temperature.*

Bronsted acid: *proton donor*

Bronsted base: *proton acceptor*

Catenation: *the tendency of an element to form long chain, branched chain and cyclic chain compounds.*

Chemical equation: *Short hand notation of a chemical reaction.*

Chemical Kinetics: *The study of the rate of a reaction and reaction mechanism.*

Density: *The ratio of the mass of substance to its volume.*

Diffusion: *The gradual mixing of the moleucles of two or more substances owing to random molecular motion.*

Electrode: *A strip of metal or grahpite for conducting electrons into and out of solutions.*

Electrolysis: *The process of converting electrical energy into chemical energy.*

Electron: *Negatively charged subatomic particle.*

Electronegativity: *The power of an atom to attract shared electrons toward itself.*

Element: *A pure substance composed of one kind of atom.*

Enthalpy Change (ΔH): *The difference between heat content of product and reactant.*

Equilibirum: *A condition in whcih the forward and reverse reaction rates in a physical or chemical system are equal.*

Funcitonal group: *Part of the compound that determines the properties of that compound.*

Graham's law: *The rate of diffusion of two gases is inversely proprotional to the square root of their molar masses or densities.*

Ground state: *The lowest energy stationary state.*

Group: *The vertical column in the priodic table.*

Halogens: *Group VIIA elements.*

Hydrogen bonding: *The strongest intermolecular force.*

Hydrogenation: *The addition of hydrogen to unsaturated compounds.*

Hygroscopic compound: *Absorbs water from the air.*

Ideal gas law: *Relates pressure, volume, temperature and number of mole of an ideal gas.*

Intermolecular forces: *Interactions between molcules, between ions, or between ions and molecules.*

Ionic Bond: *The attraction force between cations and anions.*

Isomers: *Two or more compounds with the same molecular formula but different structural formula.*

Isotopes: *Atoms of the same element with different neutron number.*

Lewis acid: *Electron pair acceptor.*

Lewis base: *Electron pair donor.*

Limiting reactant: *The reactant present in small supply that controls the amount of product.*

Mass: *The amount of matter contained in a body.*

Mass Number: *The number of nucleous in an atom.*

Molar mass: *The mass of one mole of a substance.*

Molarity: *The number of mole of solute per liter of solution.*

Orbital: *Region in space, around the nucleus where high probability of finding an electron.*

Oxidation: *The loss of electron by an atom or ion or molecule.*

Oxides: *Binary compounds containing oxygen.*

Percentage Yield: *The ratio of actual yield to theoretical yield times hundred.*

Periods: *The horizontal rows in the periodic table of elements.*

Proton: *A positively Charged sub-atomic particle found in the nucleus.*

Radioactivity: *The spontaneous emission of energy without absorption of energy.*

Reactant: *Starting material in a chemical reaction.*

Reduction: *The gain of electron by atom, ion or molecule.*

Salt: *An ionic compound formed from acid and base.*

Salt bridge: *An inverted u-tube filled with salt solution that maintains elecrical neutrality between the two electrodes solutions.*

Strong acid: *Ionizes almost completly in aqueous solution.*

Sublimation: *The direct change of solid to gas.*

Theoreticla yield: *The amount of product theoretically obtained from balanced chemical equation.*

Unsaturated hydrocarbon: *Hydrocarobn with carbon - carbon double or triple bond.*

Valence electron: *Electron that is found in the valence shell.*

Vapor Pressure: *The pressure of the vapor of a substance in contact with its liquid or solid in a sealed contianer.*

Voltality: *The tendency of a substance to vaporize at room temperature.*

PERIODIC TABLE OF THE ELEMENTS

	1 IA												13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 H Hydrogen 1.008	2 IIA																2 He Helium 4.003
2	3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
3	11 Na Sodium 22.99	12 Mg Magnesium 24.31	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
4	19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 51.99	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
5	37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Cesium 132.91	56 Ba Barium 137.33	57 La Lanthanum 138.91	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (263)	107 Bh Bohrium (264)	108 Hs Hassium (269)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (271)	111 Rg Roentgenium (272)	112 Uub Ununbium (277)						

58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.97
90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.04	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (254)	103 Lr Lawrencium (262)